

A Crisis of Banks as Liquidity Providers

VIRAL V. ACHARYA and NADA MORA*

ABSTRACT

Can banks maintain their advantage as liquidity providers when exposed to a financial crisis? While banks honored credit lines drawn by firms during the 2007 to 2009 crisis, this liquidity provision was only possible because of explicit, large support from the government and government-sponsored agencies. At the onset of the crisis, aggregate deposit inflows into banks weakened and their loan-to-deposit shortfalls widened. These patterns were pronounced at banks with greater undrawn commitments. Such banks sought to attract deposits by offering higher rates, but the resulting private funding was insufficient to cover shortfalls and they reduced new credit.

THIS PAPER INVESTIGATES WHETHER the onset of the 2007 to 2009 crisis was, in effect, a crisis of banks as liquidity providers, which increased the fragility of the financial system. The starting point of our analysis is the widely accepted notion that banks have a natural advantage in providing liquidity to businesses through credit lines and other commitments established during normal times. Even though banks experience significant and synchronized drawdowns during episodes of market stress, it is argued that they have no difficulty meeting these increased credit demands (e.g., Gatev and Strahan (2006)). The reason is that banks are awash with funds from depositors seeking a safe haven due to both deposit insurance as well as the regular occurrence of crises outside the banking system (e.g., the autumn of 1998 following the Russian default and Long-Term

*Acharya is with New York University, NBER, CEPR, and ECGI. Mora is with the Federal Reserve Bank of Kansas City. This paper previously circulated as “Are Banks Passive Liquidity Backstops? Deposit Rates and Flows during the 2007-2009 Crisis.” Jacob Schak, Paul Rotilie, Thad Sieracki, Jon Christensson, and Kristen Regehr provided valuable research assistance. We are grateful to Ruth Judson for her help with the Bank Rate Monitor data. We thank Sumit Agarwal; Heitor Almeida; David Backus; Morten Bech; Lamont Black; Francisco Covas; Bob DeYoung; Cam Harvey (the Editor); Charlie Kahn; Anil Kashyap; Bill Keeton; Michal Kowalik; Lubo Litov; Jamie McAndrews; Ouarda Merrouche; Chuck Morris; Wayne Passmore; Jack Reidhill; Joao Santos; Anthony Saunders; Philipp Schnabl; Ken Spong; Phil Strahan; an Associate Editor; anonymous referees; and participants at the American Finance Association Meeting, the Financial Management Association Meeting, the Federal Reserve’s “Day Ahead” Conference on Financial Markets and Institutions, the Chicago Fed Conference on Bank Structure and Competition, the Financial Intermediation Research Society Meeting, the Office of Financial Research, the World Bank, and the Federal Reserve Banks of New York, Philadelphia, and San Francisco for helpful comments. The views expressed herein are those of the authors and do not necessarily reflect the positions of the Federal Reserve Bank of Kansas City or the Federal Reserve System. Any remaining errors are our own.

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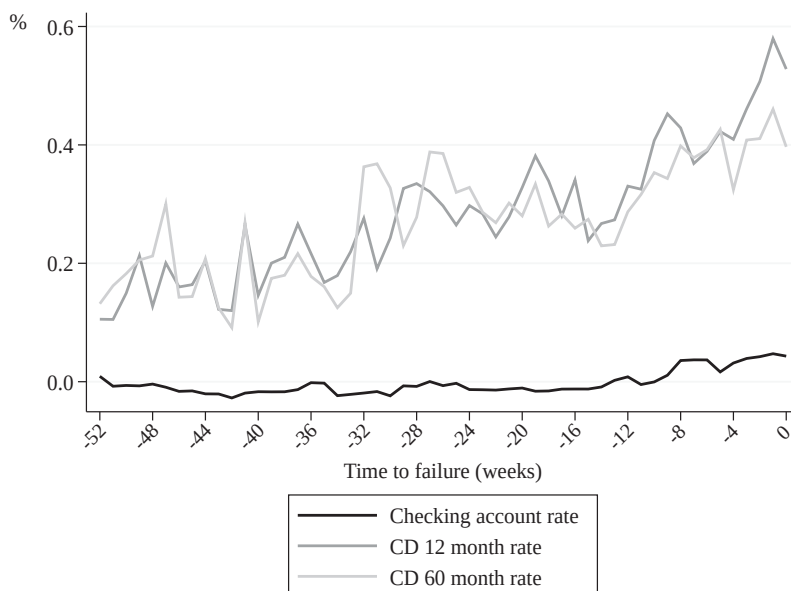


Figure 1. Deposit rates: Failed minus nonfailed banks. This figure plots the deposit rates of failed banks (and nearly failed banks) adjusted for the rates of banks that did not fail over the period 1997 to 2009. The underlying data are current rates from a weekly survey from Bank Rate Monitor (BRM), available from 1997. Specifically, the series are the average of the difference between rates for 43 failed banks (FDIC regulatory-assisted failures and CRSP near-fails) and rates of nonfailed banks. Note that, following Acharya et al. (2010), these near-fails correspond to the first date when the 18-month return was worse than -90% using daily CRSP equity data. The x -axis is the time to failure in weeks. The failure date is the last available weekly (Friday) survey date prior to failure.

Capital Management (LTCM) hedge fund failure; the 2001 Enron accounting crisis).

In 2007 to 2009, however, the banking system itself was at the center of the financial crisis. As noted by Acharya, Schnabl, and Suarez (2013), Brunnermeier (2009), and Diamond and Rajan (2009), among others, a significant portion of “toxic” financial instruments found their way onto commercial and investment bank balance sheets, raising questions about these institutions’ solvency. As the solvency risk of a bank increases, it might seek to attract deposits by offering higher rates.¹ Figure 1 shows that this has indeed been the case for the average deposit rates offered by *failed* banks and *nearly failed* banks (decline in stock price of 90% or more), as measured by the difference from the rates of *nonfailed* banks (that is, banks that did not fail) over a one-year period prior to failure for failures occurring during the 1997 to 2009 period (concentrated within the crisis). For ease of comparison, the x -axis gives the time to failure.

¹ For example, Washington Mutual pitched above-market rates prior to its acquisition by JP Morgan Chase in 2008 as did Countrywide in 2007. See the *American Banker*, October 9, 2008, “Deposit-Gathering Pitches Evolving Amid Upheaval” and December 5, 2007, “Why Rate Cuts Aren’t Helping on the Deposit Side.”

As is evident, weak institutions offered substantially higher CD rates in the run-up to failure.²

This rise in deposit rates reflected stress on the deposit funding of banks. In particular, not all deposit liabilities were insured, especially at the onset of the crisis when over 62% of deposits were uninsured. Did banks manage to retain their deposit base in the wake of the funding stress? And did they manage to meet their credit demands in the form of undrawn credit lines and commitments to firms?

Our main result is that, until the government interventions in the fall of 2008, the mechanism whereby the banking system as a whole provides backup liquidity to the market by experiencing deposit inflows broke down. This crisis was in fact a crisis of banks as liquidity providers in the aggregate, not just of the weakest banks. In particular, deposit funding pressure was widespread and particularly acute in the first phase of the crisis from the asset-backed commercial paper (ABCP) “freeze” starting August 9, 2007 (as documented in Acharya, Schnabl, and Suarez (2013)) until just before the Lehman failure on September 15, 2008. Core deposits increased by only \$90 billion up until end-2008Q2 (an increase that did not exceed core deposit inflows in a comparable period just before the crisis).³ Core deposits eventually increased in the banking system as a whole by close to \$800 billion by early 2009, but only starting in 2008Q3, when they grew by \$272 billion in just one quarter (Table I and He, Khang, and Krishnamurthy (2010)). Importantly, prior to Lehman’s failure, lending growth outpaced core deposit funding growth, that is, at the aggregate level, the banking system recorded a loan-to-deposit shortfall throughout the first year of the crisis.⁴

We argue that the weakness in the aggregate deposit funding position of banks and its sharp reversal following Lehman’s failure is explained by investor perception of greater risk in bank deposits relative to instruments offering similar liquidity and payments services. Because most deposits were

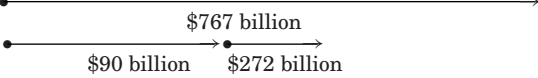
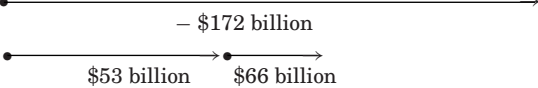
² This finding conforms to a well-developed strand of the market discipline literature that shows various types of bank claim holders are able to identify risky banks in a timely manner and penalize them by withdrawing funds and raising the costs of additional funds. Flannery (1998) provides a good review of the capacity of such market mechanisms to supervise and remedy excessive risk-taking by financial firms, and thus we do not dwell in detail on these results.

³ For example, the \$90 billion increase in core deposits at all banks in the first half of 2008 fell short of an average \$130 billion increase in the preceding five years. The gap was even greater for small banks (\$7 billion compared with \$40 billion in the preceding five years) than for large banks (\$82 billion compared with \$90 billion in the preceding five years). The weak behavior of core deposit growth is striking in light of the substantial easing of monetary policy during the 2007 to 2009 crisis. From September 2007 to April 2008, the federal funds target was reduced by 3.25 percentage points, and the three-month Treasury bill rate fell nearly three points. Such a reduction in market interest rates (the opportunity cost of deposits) usually leads to faster growth of core deposits. Similar evidence of initially weak deposit growth and the contrast with 1998 is documented in Ashcraft, Bech, and Frame (2010).

⁴ Note that the application of terms such as “pressure” or “squeeze” to aggregate deposits in this paper is best interpreted as the *relation* of deposit funding to loan demand and drawdowns. That is, characterizing deposit changes in isolation is insufficient to address whether the banking sector functioned adequately in its natural liquidity hedge role. We examine this question in more detail in Section I.B.

Table I
Cumulative Deposit Growth during the 2007 to 2009
Financial Crisis (in %)

This table presents an extract of cumulative growth statistics for certain deposit accounts based on the quarterly Call Reports over the period from 2007Q3 to 2009Q1 (see the online version of this article for full balance sheet details). Panel A shows the growth of insured deposits, core deposits, and large time deposits over the period relative to initial 2007Q2 levels for large banks. Similarly, Panel B shows the deposit growth figures for small banks, and the bottom panel summarizes overall deposit inflows into the banking system in billions of dollars over the period 2007Q4 to 2009Q1. These aggregate flow of funds match the data reported in He, Khang, and Krishnamurthy (2010) for the corresponding period (where core deposits are reported to have increased by roughly \$800 billion while large time deposits decreased by \$200 billion).

	2007Q3	2007Q4	2008Q1	2008Q2	2008Q3	2008Q4	2009Q1
Panel A: Large Banks (Largest 25 Banks, H8 Criteria)							
Insured deposits	0.1	0.8	1.7	1.7	3.9	4.6	5.6
Core deposits	0.2	2.4	3.8	3.4	6.3	9.7	10.8
Large time deposits	0.6	1.5	1.8	1.3	2.8	2.0	1.2
Panel B: Small Banks							
Insured deposits	0.0	0.0	0.7	1.0	2.3	3.0	2.0
Core deposits	−0.4	−0.7	−0.7	−0.5	0.0	0.9	0.5
Large time deposits	1.7	2.4	3.4	3.8	2.7	−0.2	−0.1
All Banks, Core							
All Banks, Large Time							

over the deposit insurance limit at the outset, investors preferred to hold assets with an explicit government guarantee such as Federal Home Loan Bank (FHLB) discount notes (analyzed in Ashcraft, Bech, and Frame (2010)) and Treasury securities (directly and indirectly through money market mutual funds (MMMFs) specializing in government securities). The funding inflow into government funds exceeded that into “prime” funds beginning in August 2007,⁵ and accelerated in the aftermath of Lehman’s failure, the Reserve Primary Fund’s “breaking the buck,” and the sharp outflow from prime funds. Concurrently, the government backed the depository system more explicitly through an increase in insurance limits from \$100,000 to \$250,000 and the full insurance of noninterest-bearing accounts, among other measures. Therefore, explicit government backing appears to have been the key factor explaining the aggregate funding shifts.

To understand the microeconomics of these aggregate effects on the banking sector, we analyze the liquidity provision role of banks at the individual

⁵ Prime funds invest in privately issued short-term, highly rated, liquid securities.

bank level. Our results show that the aggregate liquidity shock at the onset of the crisis particularly hit banks exposed to drawdowns of commitments and credit lines. These banks were vulnerable to liquidity risk because liquidity tensions coincided on both sides of their balance sheet in the crisis. Drawing on evidence from quarterly Call Reports and CD rates from a weekly proprietary survey, we find that a bank exposed to high commitments increased deposit rates significantly more than one exposed to low commitments (regardless of whether exposure is measured by potential or effective drawdowns). But crucially, despite scrambling for deposits by raising rates, commitments-exposed banks experienced weaker deposit growth, including of core deposits that are commonly considered a stable source of funding, and were forced to cut back on new credit originations.⁶

The credit adjustment of banks exposed to liquidity risk in the form of undrawn commitments might seem at odds with existing evidence from firm-level data (based on surveys and SEC filings) in Ivashina and Scharfstein (2010), Campello et al. (2011), and Berrospide, Meisenzahl, and Sullivan (2012), among others, which shows that firms drew heavily on credit lines with banks beginning in August 2007, and that banks honored their *ex ante* promises and met this demand. They conclude that corporate investment was cushioned because of credit line availability as cancellations by banks were rare. We show, however, that this provision of liquidity insurance by banks was possible *only* because of explicit, large support from the government and government-sponsored agencies (FHLB advances and Federal Reserve liquidity facilities). For example, advances from the FHLBs covered 65% of nondeposit borrowing growth at commitments-exposed banks during the first year, and the widening shortfall between their on-balance-sheet loans and deposits was closed halfway with government-sponsored borrowing.⁷ Equally importantly, we document that, unlike previous crises, banks did not expand total loans and credit lines. In this sense, the role of banks as liquidity providers was itself in crisis during the 2007 to 2008 period of the crisis.

We conduct several tests to rule out the alternative hypothesis that commitments-exposed banks were simply those with greater solvency problems, and thus expected to be disciplined by depositors, as discussed in footnote 2. The results indicate that solvency problems, such as exposure related to real estate, were relevant risk factors whose effect persisted (as expected) even after

⁶ While they experienced a positive inflow of insured deposits, this was statistically insignificant and economically small during the first year of the crisis. These banks also were more likely to seek expensive brokered deposits and to pay a fee to participate in the FDIC's transaction account guarantee (TAG) program, which was established in October 2008 to guarantee noninterest-bearing transaction accounts. Moreover, reintermediation through interbank markets to these liquidity-short banks was insufficient. It only bridged roughly one-quarter of their total nondeposit borrowing growth during the first year.

⁷ Indicative of the extent to which the overall U.S.-chartered commercial banking system was reliant on federal sources of funds, Federal Reserve loans rose \$559 billion from 2007Q4 to 2008Q4, compared with a total net increase in bank liabilities of \$1,421 billion. The FHLB stepped in earlier, with its loans increasing \$261 billion from 2007Q3 to 2008Q3 (Federal Reserve's Flow of Funds Table F.110 for U.S.-chartered commercial banks).

the deposit funding squeeze eased in the latter part of the crisis. In contrast, the funding pressure on commitments-exposed banks coincided with the shifts in aggregate deposit funding. The sharp liquidity reversal following Lehman's failure due to explicit government guarantees benefited banks with promised outstanding lines of credit and other liquidity-exposed banks (such as banks dependent on wholesale funding). We also show that commitments-exposed banks with weaker fundamentals were more vulnerable to the onset of the crisis than equally exposed banks with stronger fundamentals. These findings are motivated by theoretical models (e.g., Rochet and Vives (2004), Diamond and Rajan (2005)) in which there is a range of fundamentals for which a solvent bank can be illiquid. Interestingly, however, even the healthier commitments-exposed banks did not experience a funding inflow in the first year of the crisis and therefore were not well placed to reintermediate funds within the banking system.

The rest of the paper is organized as follows. Section I begins with a brief review of the theory that banks can provide liquidity when financial markets and other financial institutions cannot, and discusses why the theory might break down in a financial crisis. We support this discussion with aggregate evidence from the recent crisis. Section II investigates how exposure to heightened liquidity demand risk at the individual bank level shaped deposit rates and was reflected in funding flows and lending shortfalls. Section III addresses the robustness of the results and reconciles our findings with previous crises. Section IV concludes with policy implications.

I. Banks as Liquidity Providers: The Traditional View and Its Reconsideration

A. The Rationale for Banks as Liquidity Providers

Banks are considered as having an important advantage over other financial institutions in providing various forms of liquidity commitments such as corporate lines of credit and demandable deposits. This advantage relates to how they resolve the liquidity management problem that arises when commitments are converted into funded loans and immediacy is demanded on deposits. As described by Kashyap, Rajan, and Stein (2002), banks combine deposit-taking with loan commitments. A synergy exists between these two activities to the extent that both services require banks to hold balances of liquid assets to provide liquidity on demand to depositors as well as to credit line borrowers. In particular, banks have a natural advantage in providing liquidity, provided deposit withdrawals and commitment drawdowns are not too highly correlated.

Indeed, studies have shown that, during past episodes of market stress, deposit withdrawals and commitment drawdowns were *negatively* related (Saidenberg and Strahan (1999), Gatev and Strahan (2006), Gatev, Schuermann, and Strahan (2009)). For example, when the commercial paper market encountered stress, the funds that investors pulled out of this market flowed primarily into the banking system just as borrowers resorted

to drawing down their bank credit lines. These studies argue that investors withdrew funds from markets at large, not just the commercial paper market, and deposited their funds with the banking system because it was seen as a safe haven given sufficient government guarantees on deposits.⁸ The view that government guarantees are responsible for the deposit inflows during crises is supported by evidence that no such deposit inflows occurred prior to the FDIC's inception in 1934. Pennacchi (2006) shows that, during times of tight market liquidity from 1920 to 1933, no increase in bank deposits was observed and liquidity shocks were accompanied by declines in bank loans and investments. Interestingly, U.S. banks did not extend formal loan commitments in the pre-FDIC era.

B. When the Liquidity Backup Mechanism of the Banking System as a Whole Breaks Down

While it may make sense *ex ante* for banks to combine deposit-taking with loan commitments when liquidity demand realizations of depositors and firms are idiosyncratic or not very positively correlated, banks may experience *ex post* a coincident liquidity demand from depositors and firms, leading to a more fragile financial system. Banks might be forced to run down liquid assets and borrow from outside the interbank system to make good on promised loans. They likely will be forced to adjust to the shock by cutting back on new lending. This section provides supporting evidence from the U.S. subprime crisis in 2007 to 2008.

At the outset of the U.S. subprime crisis, investors (households) seeking safety piled into securities issued or sponsored by the government, not deposits and debt issued by banks. Some argue that the initial subprime shock and lack of information about bank exposures to the shock led investors to lose confidence in their ability to identify low- from high-risk banks, leading to a collective withdrawal from deposit accounts (e.g., Gorton (2008), Covitz, Liang, and Suarez (2013)).⁹ Others argue that, when aggregate risk rises, the ability

⁸ One point of departure from Kashyap, Rajan, and Stein (2002) that Gatev and Strahan (2006) take is on what drives deposit flows in times of reduced market liquidity. The thesis in Kashyap, Rajan, and Stein is that the coexistence of deposits and lending is due to real benefits and is not the result of historical or contemporary regulatory distortions. That is, they do not think it is an artificial synergy resulting from deposit insurance. Gatev and Strahan reason instead that "banks can insure firms against systematic declines in liquidity at lower cost than other institutions" precisely because of a general flight to safety, triggering inflows into the banking system (p. 867). They conjecture that this mechanism is largely due to government support, both explicit through deposit insurance and implicit through expected bailouts. The two views are not necessarily orthogonal, however.

⁹ Indeed, Rajan (2005) was prescient about the transmission of the 2007 to 2009 crisis. He argued that the reason banks were able to honor their credit lines when the commercial paper market dried up in 1998 was that banks were not perceived as credit risks themselves so that funds flowed into banks. But he went on to say that if "banks also face credit losses and there is uncertainty about where those losses are located, only the very few unimpeachable banks will receive the supply of liquidity fleeing other markets. If these banks also lose confidence in their

of banks to diversify or smooth shocks across corporations and depositors is disrupted (Acharya, Almeida, and Campello (2013)). In anticipation, banks may raise costs for provision of liquidity insurance, for example, increase spreads on corporate lines of credit. In response to reduced bank liquidity, depositors may leave banks, especially those that are heavily exposed to the rise in aggregate risk and the underlying economic shock. For this argument to hold, it is necessary that a significant share of deposits not be explicitly guaranteed by the government (which was true since most deposits in the banking system were over the deposit insurance limit at the onset of the crisis; more than 62% in 2007Q2 when the limit was only \$100,000). Lacking an explicit guarantee,¹⁰ investors preferred to hold assets with explicit guarantees, such as FHLB discount notes (analyzed in Ashcraft, Bech, and Frame (2010)) and Treasuries.

One main asset through which investors hold Treasuries is through MMMFs specializing in government securities. Generally, MMMFs (both government and prime funds) are seen as competing deposit collectors because they offer similar liquidity and payments services (Pennacchi (2006)). Investors also saw MMMFs as safe up until the Lehman bankruptcy and the ensuing losses and redemption demands on the Reserve Primary Fund and other funds in September 2008. Importantly, however, while both government and so-called prime funds received large inflows beginning with the subprime crisis, the percentage increase was significantly *greater* at funds devoted to government securities even prior to Lehman's failure (Figure 2, Panel A). For example, from August 2007 to pre-Lehman's failure, the change in government (prime) funds was \$465 billion (\$411 billion), the detrended change in government (prime) funds was \$422 billion (\$182 billion), and the percentage change in government (prime) funds was 101% (25%). This occurred despite a jump in the yield on prime funds relative to government funds (Figure 2, Panel B).

The inflow into government funds gathered pace in the aftermath of Lehman's failure and the Reserve Primary Fund's net asset value falling below one dollar when there was a sharp outflow from prime funds.¹¹ For example, in the two weeks following Lehman's failure, prime funds lost \$410 billion and government funds gained \$238 billion, partly reallocated from prime funds. Assets in prime funds stabilized end-September 2008 only after several government

liquidity-short brethren, the interbank market could freeze up, and one could well have a full blown financial crisis."

¹⁰ This lack of guarantee can become especially important in a bank-centered crisis, when uncertainty about the condition of an individual bank may prompt depositors to run from the bank even if it turns out to be ex post fundamentally solvent. For example, Iyer and Puri (2012) show that depositors ran from a fundamentally sound bank when a nearby bank failed and that deposit insurance was only partly successful at limiting outflows of insured deposits.

¹¹ The Reserve Primary Fund had close to an \$800 million exposure to Lehman commercial paper. While only the Reserve Primary Fund halted redemptions on September 16, 2008, other funds were close to halting redemptions. For example, a number of troubled funds (such as Columbia, Dreyfus, General, ING, and Morgan Stanley) filed petitions with the SEC, which in turn issued no-action letters, indicating that the sponsor provided financial support to its fund (Kacperczyk and Schnabl (2013)).

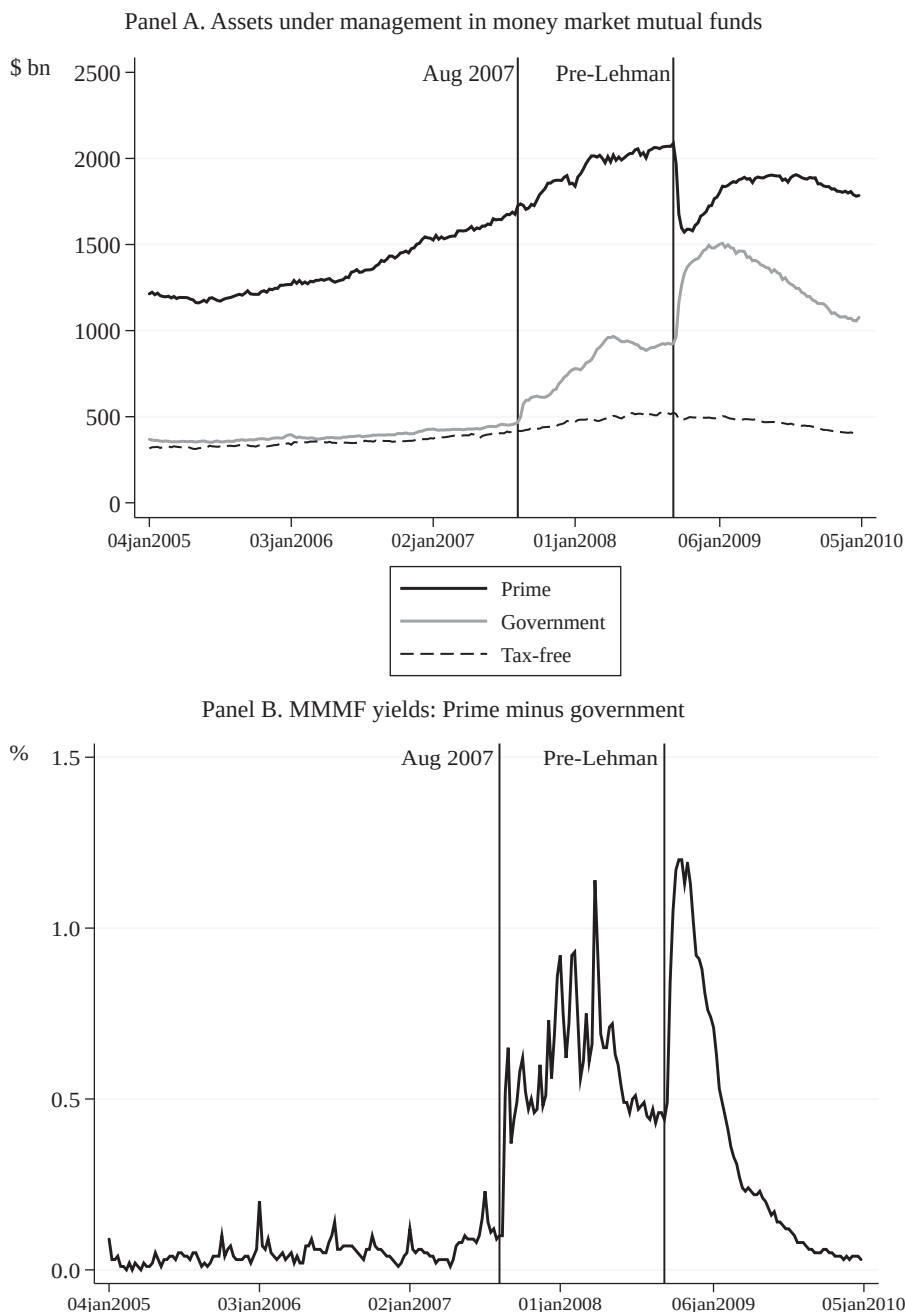


Figure 2. Money market mutual funds. Panel A plots assets under management in money market mutual funds (MMMFs) by investment type. Panel B plots the net yield for prime funds minus that for government funds. The sample is weekly between 2005 and 2009. The data source is iMoneyNet for MMMFs, weekly data.

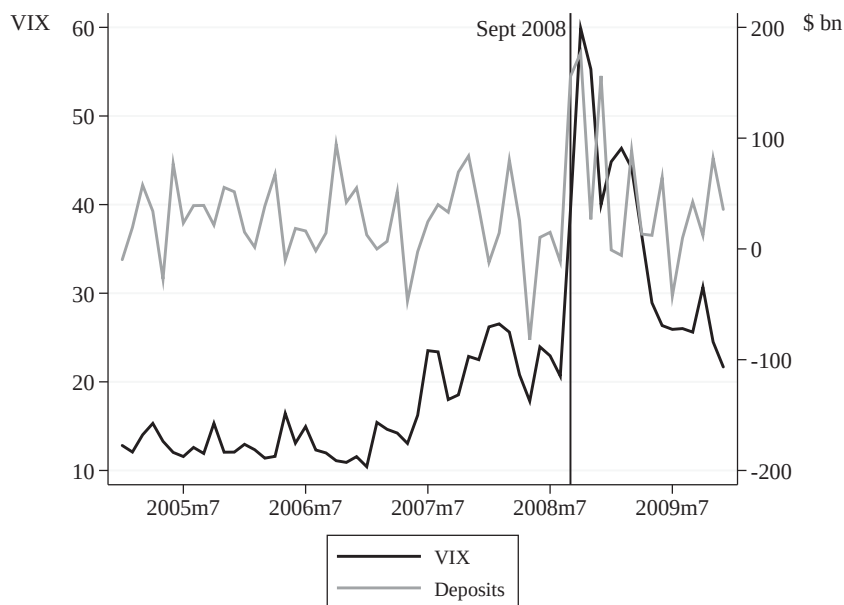


Figure 3. Market stress and net flows into deposits at commercial banks. This figure plots the VIX index (left axis) and net flows into deposit accounts at commercial banks (right axis) over the period 2005 to 2009. The underlying data are the Chicago Board Options Exchange for VIX (month-end), and the Federal Reserve H8 release for deposits at domestically chartered commercial banks. The latter deposit flows data are aggregated from the weekly to monthly frequency. These deposit flows are also corrected for the artificial impact of JP Morgan Chase's acquisition of the large thrift WaMu reflected in the H8 release of October 1, 2008.

actions including new Federal Reserve liquidity facilities (such as the ABCP Money Market Mutual Fund Liquidity Facility (AMLF) and the Commercial Paper Funding Facility (CPFF)) and the Treasury's introduction of temporary insurance for all MMMF investments made prior to Lehman's failure (Adrian, Kimbrough, and Marchioni (2010), Kacperczyk and Schnabl (2010)).

At the same time, deposits in the banking system shot up by \$189 billion. This followed a period of anemic inflow of deposits in the banking system from the ABCP freeze in the summer of 2007 to just before the Lehman failure (Figure 3 and Table II) (the surge was visible across both core and large time deposits and at both large and small banks). The absence of a sharp deposit inflow at the onset of the crisis is surprising when compared with the behavior in previous crises when commercial paper spreads also exceeded 100 bps. For example, Table II illustrates the robust deposit inflow in the fall of 1998 (more than quadrupling to \$8.7 billion per week, equivalent to \$100 billion by year-end) and its absence in 2007 to 2008. A similar outward shift in the supply of funding to banks (by \$200 billion) followed the 9/11 attacks as market liquidity dried up and commercial paper spreads rose. Therefore, the flight-to-safety to the banking system that systematically occurred previously broke down early in this crisis and was not observed until September 2008.

Table II

Average Weekly Deposit Funding and Lending Changes (in \$ Billions)

This table reports the average weekly deposit funding and lending changes that occurred in the 2007 to 2009 crisis and compares these changes to those that occurred during the 1998 LTCM Crisis. The sample is weekly from the Federal Reserve H8 Release for all reporting domestically chartered banks.

	1998 LTCM Crisis		2007 to 2009 Financial Crisis			
	Previous Five Years to Crisis	Fall 1998	Previous Five Years to Crisis	July 4, 2007, to Sept 10, 2008 (Pre- Lehman Failure)	Period Immediately Following Lehman Failure	Oct 8, 2008, to July 1, 2009
Deposits	2.1	8.7	6.6	6.5	94.7	9.0
of which, core deposits	1.4	7.7	5.2	4.7	59.2	12.2
Loans	3.5	8.1	8.1	9.5	38.2	-5.3
of which, C&I loans	1.0	2.1	0.9	3.1	15.8	-2.5
Loan-to- deposit shortfall	1.4	-0.5	1.6	3.0	-56.5	-14.3
of which, small banks	0.4	-0.4	0.2	1.6	-4.0	-6.1

This aggregate funding shift following Lehman's failure is explained by both the acute flight-to-safety out of prime funds because these were suddenly perceived to be even riskier than bank deposit accounts, and the concurrent backing of the banking system made by the government. For example, in addition to recapitalization measures, the Emergency Economic Stabilization Act (authorizing the Troubled Asset Relief Program (TARP)) increased the deposit insurance limit from \$100,000 to \$250,000 per depositor. This change was followed by the FDIC's announcement of its temporary liquidity guarantee program, in which it stood willing to guarantee newly issued senior unsecured debt of banks and thrifts and fully cover noninterest-bearing deposit transaction accounts, largely held by businesses (known as the TAG program). Together with other implicit guarantees of the financial system, these measures assured depositors that the banking system would hold up. As a result, deposits poured into banks.¹²

¹² Absent increased government support of the banking system, we would not expect that much of the outflows from prime funds found their way into a wide range of banks and deposit accounts, not just the safest of them. Moreover, Afonso, Kovner, and Schoar (2011) observe that, once government support of AIG was announced two days after Lehman's failure, interbank spreads returned to their precrisis levels. This effectively signaled a backstop even before TARP was finally passed by Congress on October 3, 2008. Other factors contributed to weak deposit inflows through the crisis but these do not explain the shift following Lehman's failure. Reasons include (i) the hit to household wealth (stock and property markets continued to decline), (ii) the withdrawal of

The inflow of deposits finally allowed the banking system as a whole to close its loan-to-deposit shortfall that had opened up during the first year of the crisis (Table II and Figure 4). The shortfall refers to the difference between lending growth and deposit growth, where growth in Figure 4 is measured by the cumulative growth from the beginning of the crisis in July 2007. For example, the shortfall widened to \$198.1 billion by the end of 2007Q3, reached a cumulative \$239.2 billion by 2008Q2, and rose to over \$300 billion in the weeks prior to Lehman's failure, based on snapshots of the weekly data. On-balance-sheet lending increased, reflecting loan commitment takedowns and mortgage loans that were warehoused because they could no longer be securitized. Other nondeposit noninterbank borrowing helped to support lending growth (see the Internet Appendix).¹³ Banks, especially small banks, ran down their most liquid assets to support lending and the buildup of assets (reflected in cash assets and Treasury and agency securities; see the Internet Appendix).

The next section tests whether this shortfall seen in the aggregate banking system was indeed a problem for banks exposed to liquidity demand risk. That is, was the role of banks as liquidity providers itself in crisis during the first year of the crisis?

II. Liquidity Demand Risk and Individual Bank Behavior in the 2007 to 2009 Crisis

A. Hypotheses

As we discuss in the previous section, at the outset of the crisis the banking system as a whole was short of deposits relative to lending and other asset funding needs. This section begins by developing testable hypotheses for individual bank behavior when banks need funds to support increased loan demand but face deposit funding pressure. Next, we test whether commitments-exposed banks were liquidity-short, and if so, whether they promised higher rates in an effort to stem deposit outflows and what additional actions, if any, they took to adjust their balance sheets to the shock. There is considerable variation in a bank's vulnerability to the risk of liquidity demand, controlling for its wholesale funding, solvency measures, and size. We therefore expect that a bank at greater risk of credit line drawdowns offers higher rates during the crisis

deposits by liquidity-constrained households to cover consumption needs (also continued), and (iii) the FDIC's negative reserve balance that may have led depositors to worry that the FDIC would not provide enough resources to make them whole in the event of failure (also persisted; the Treasury later increased its line of credit to the FDIC in March 2009).

¹³ The Internet Appendix may be found in the online version of this article. We show that borrowing from "other" (i.e., non-U.S. banks) dwarfed borrowing from U.S. banks, and that large banks also pulled in funds from related foreign offices (consistent with Cetorelli and Goldberg (2012)).

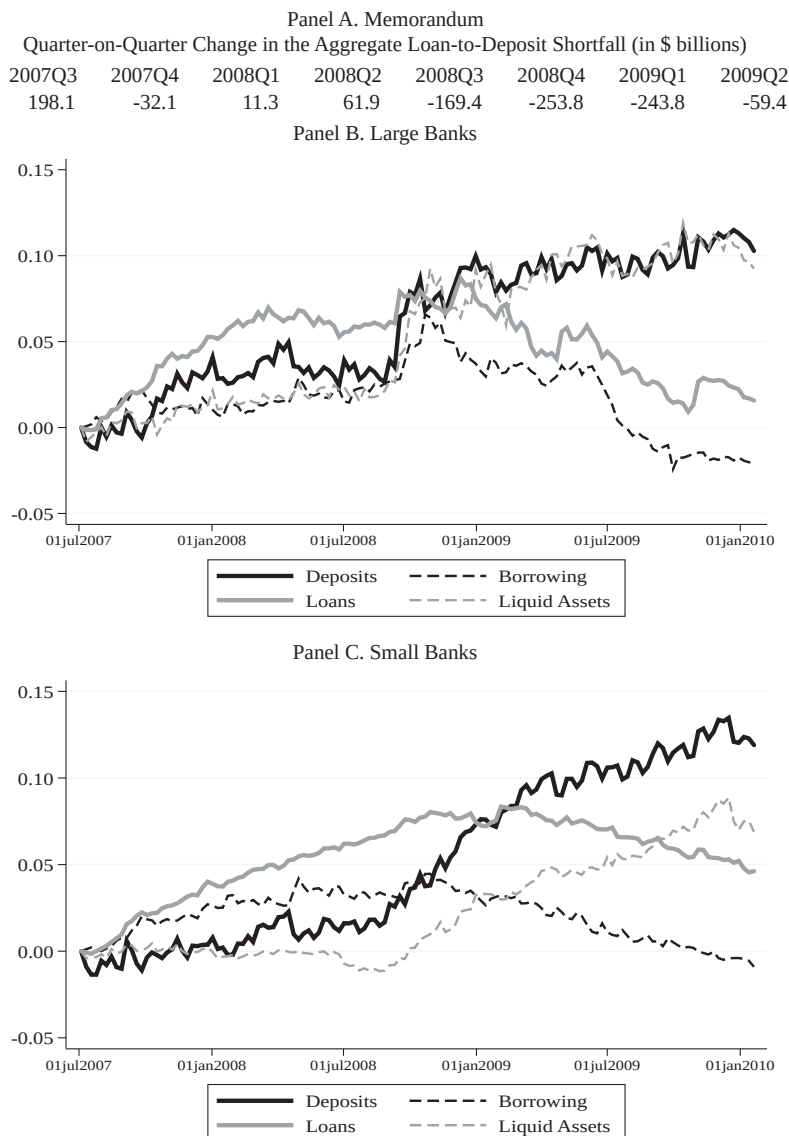


Figure 4. Cumulative growth of balance sheet components. Panel A shows quarterly snapshots of the loan-to-deposit shortfall for all domestically chartered banks based on the underlying weekly data. Panels B and C plot the cumulative growth of key balance sheet assets and liabilities at the weekly frequency from July 2007 through the end of 2009 (growth figures are relative to the first week of July 2007). The four key series are total deposits, nondeposit borrowing, loans and leases, and liquid assets (the sum of cash assets, securities, and interbank loans extended). Panel B is for large domestically chartered banks and Panel C is for small domestically chartered banks. The underlying data are the Federal Reserve H8 release for deposits at domestically chartered commercial banks. Large banks in the H8 release are the largest 25. For example, inclusion as of mid-2009 was \$65 billion asset size. Combined, large banks held 66% of domestically chartered assets and 60% of deposits. The balance sheet series at the large banks are also corrected for the artificial impact of JP Morgan's acquisition of the large thrift WaMu reflected in the H8 release of October 1, 2008.

if it does not gain (or anticipate gaining) proportionate deposits to match its funding needs. We estimate the model

$$\begin{aligned} \text{Deposit Rate}_{i,t} = & \beta_1 \text{Liquidity Demand Risk}_{i,t-1} \\ & + \beta_2 \text{Liquidity Risk}_{i,t-1} \times \text{Crisis}_t + \text{Bank Fixed Effect}_i \quad (1) \\ & + \text{Time Fixed Effect}_t + \text{Other Controls}_{i,t} + e_{i,t}, \end{aligned}$$

where *Other Controls* captures time-varying changes in a bank's local geographic market conditions such as deposit market concentration and district time trends.

Because our thesis is predicated on the reversal in the aggregate liquidity shock in the latter half of the financial crisis, the dummy variable *Crisis* is empirically represented by two dummy variables, *Crisis1* and *Crisis2*. The variable *Crisis1* refers to the period from 2007Q3 to 2008Q2, while *Crisis2* refers to the period from 2008Q3 to 2009Q2. This setup allows us to test whether bank behavior differed in the early and late stages of the crisis, as represented by the interaction terms between bank fundamentals and *Crisis1* or *Crisis2*. We test whether a commitments-exposed bank actively responded to the deteriorating liquidity position at the onset of the crisis by offering higher rates, and then with the reversal of the banking system's position to a liquidity-rich one, this bank lowered its rate. For example, Gatev and Strahan (2006) show that a funding advantage for commitments-exposed banks during periods of heavy inflows to the banking system takes the form of consistently higher flows into deposit accounts at these banks and lower yields on their deposits.¹⁴ We also conduct several tests on the timing of the liquidity reversal, including exploiting a weekly proprietary survey of CD rates.

In addition to tests of deposit rate actions, we test whether banks took additional actions to meet increased loan commitment demand. Absent sufficient deposit insurance to ensure adequate core deposit funding and absent sufficient reintermediation through interbank markets to liquidity-constrained banks (we test for both claims), a bank will likely be forced to adjust to a liquidity shock by cutting back on new credit. Other backup actions include running down liquid assets and seeking out government and government-sponsored borrowing. We test whether commitments-exposed banks took these possible actions by specifying models similar to equation (1) in which the dependent variable is replaced with a bank's growth of borrowing from government-sponsored agencies, for example.

B. Data and Variable Description

Table III presents summary statistics for the main variables; a detailed description of the variables is available in the Internet Appendix. Most

¹⁴ The possibility that these banks anticipate getting inflows during periods of market stress and therefore offer lines of credit ex ante is a likely explanation. For example, banks made little, if any, term loan commitments and lines of credit in the pre-FDIC era. An interesting question is whether the current significantly lower commitment lending than the precrisis ratio marks a

Table III
Summary Statistics for Regression Bank Panel, 1990Q1 to 2009Q4

This table presents summary statistics for the main regression variables. These statistics are calculated over the regression sample (and thus exclude mergers, non-U.S.-domiciled banking organizations, and those below \$100 million in total assets). See the Internet Appendix for variable definitions and sources.

	Mean	Standard Deviation	25 th Percentile	Median	75 th Percentile	Obs.
Dependent variables (Call Reports)						
Interest rate, large time deposits (implicit, % annual)	4.526	1.571	3.344	4.567	5.538	238,557
Interest rate, core deposits (implicit, % annual)	3.406	1.443	2.265	3.397	4.295	238,543
Quarterly growth of deposits	0.011	0.031	-0.006	0.008	0.028	253,492
Quarterly growth of core deposits	0.008	0.028	-0.006	0.006	0.023	253,492
Quarterly growth of brokered deposits	0.001	0.008	0.000	0.000	0.000	236,763
Transaction account guarantee program deposit share (2008Q4)	0.033	0.045	0.004	0.021	0.044	21,674
Quarterly growth of loans	0.010	0.029	-0.003	0.010	0.025	253,492
Quarterly growth of C&I loans	0.001	0.013	-0.003	0.001	0.006	253,492
Quarterly growth of credit (loans + commitments)	0.011	0.032	-0.004	0.010	0.027	253,492
Loan-deposit shortfall	-0.154	0.240	-0.290	-0.176	-0.057	253,492
Quarterly growth of loan-deposit shortfall	0.000	0.038	-0.021	0.000	0.021	253,492
Quarterly growth of federal funds purchased and repo borrowing	0.000	0.013	0.000	0.000	0.001	253,492
Quarterly growth of other borrowed money (total)	0.001	0.016	0.000	0.000	0.001	210,745
Dependent variables (Bank Rate Monitor 1997 to 2009)						
Interest checking rate (%)	0.610	0.601	0.150	0.400	1.000	107,466
CD 12-month rate (%)	3.170	1.540	1.740	3.200	4.500	107,298
CD 24-month rate (%)	3.448	1.412	2.230	3.444	4.640	104,874
CD 60-month rate (%)	4.023	1.153	3.200	4.020	4.890	97,235
Covariates						
Unused commitment ratio	0.127	0.087	0.070	0.117	0.170	257,078
Liquidity ratio (excludes MBS/ABS)	0.242	0.144	0.141	0.218	0.312	215,100
Wholesale funding ratio	0.198	0.144	0.103	0.171	0.255	258,108
Net wholesale funding ratio (wholesale - liquid)	-0.030	0.219	-0.163	-0.031	0.099	215,100
Nonperforming loans to loans	0.012	0.020	0.002	0.007	0.014	228,042
Capital ratio (book capital to assets)	0.091	0.045	0.076	0.089	0.106	258,108
Indicator for large banks	0.007	0.084	0.000	0.000	0.000	258,108
Real estate loan share	0.642	0.235	0.527	0.682	0.809	257,015

(Continued)

Table III—Continued

Pairwise Correlation Coefficients for Key Covariates in Regression Bank Panel 1990Q1 to 2009Q4

	1	2	3	4	5	6
1 Unused commitment ratio	1.000					
2 Net wholesale funding ratio	0.053	1.000				
3 Nonperforming loans to loans	−0.036	0.044	1.000			
4 Capital ratio (book capital to assets)	0.059	−0.214	0.053	1.000		
5 Indicator for large banks	0.225	0.087	0.009	0.003	1.000	
6 Real estate loan share	0.045	0.007	0.074	0.272	−0.086	1.000

bank-level variables are from the quarterly Reports of Condition and Income that banks file with regulators (Call Reports). Banks belonging to a common holding company are aggregated to the top holder and treated as a single banking organization (following Kashyap, Rajan, and Stein (2002), Gatev and Strahan (2006)). Therefore, any reference to a “bank” in the remainder of this discussion should be taken to mean a banking organization, and standard errors are clustered at the banking organization level. The panel of roughly 7,000 banking organizations is unbalanced and merger effects are controlled for by excluding observations when the quarterly growth rate of a bank’s total assets exceeds a certain threshold—in this case, 10% (e.g., Gatev and Strahan (2006)). The sample excludes the smallest banks with assets less than \$100 million. Growth rates as well as implicit deposit rates calculated from the Call Reports are winsorized at the 1st and 99th percentiles to mitigate the effect of outliers. The regressions include fixed effects for both banks and time (quarterly time dummies), although we also carry out robustness checks on a pooled sample without bank fixed effects.

The primary deposit rates used in the analysis are the rates on large time deposits and the rates on core deposits. These are implicit rates based on quarterly Call Report data (expressed in annual percentage terms), defined as interest expenses on deposits divided by the quarterly average of the respective interest-bearing deposits.

Exposure to liquidity demand is measured by a bank’s unused commitments ratio, which is defined as the ratio of unused loan commitments to the sum of loans and unused commitments.¹⁵ Unused loan commitments are the parts of

permanent downward shift in the backup liquidity mechanism provided by the banking system to the nonfinancial sector (see the Internet Appendix).

¹⁵ Results are similar if unused loan commitments are scaled instead by the sum of assets and unused commitments. The main measure applied in this paper follows that in Kashyap, Rajan, and Stein (2002) and Gatev and Strahan (2006), and therefore excludes credit card commitments because a large share of credit card lines is unlikely to be drawn. In robustness checks, we also employ a narrower measure of unused commitments to proxy for commercial and industrial loan commitments and also check that the analysis is robust to excluding the conduit banks.

credit lines that have not been drawn down, and include, for example, support to ABCP conduits that the bank does not consolidate on its balance sheet (as described in the notes to schedule RC-L). The rationale for measuring a bank's liquidity demand exposure by the unused commitment ratio is that the more commitments a bank has outstanding, the more exposed it will be to their drawdown when funding conditions are stressed. But the key finding of previous studies is that more exposed banks are not, in fact, more fragile but instead serve as liquidity backstops. While these banks convert a lot of the off-balance-sheet commitments to on-balance-sheet loans, they experience a more than offsetting inflow of funds into deposit accounts in times of market stress. The aggregate evidence presented so far casts doubt on this liquidity insurance notion extending to the subprime crisis.

We control for a bank's other liquidity and solvency measures including net wholesale funding, nonperforming loans, capital, real estate exposure, and size. In brief, net wholesale funds are liabilities excluding core deposits less liquid assets. This variable therefore measures net wholesale borrowing including gross federal funds purchased less gross federal funds sold and repos less reverse repos. Nonperforming loans are loans 90 days past due and nonaccruing loans. Capital is measured by the book capital-to-asset ratio. Real estate exposure is mainly measured by the share of loans secured by real estate in a bank's loan portfolio (we also conduct robustness checks with a range of other real estate-related securities and trading exposure). Finally, perceived solvency risk is driven by implicit guarantees of certain banks. For example, large banks may be big enough to be considered "too-big-to-fail" (TBTF) by investors. Large banks are captured in the regressions by an indicator for the 25 largest banks as measured by asset size.¹⁶

The main sample period is quarterly from 1994 to 2009 (unless otherwise noted). Unused commitments are available from 1990 but the control for deposit market concentration is from the FDIC's Summary of Deposits, available from 1994. In addition, liquid assets are defined to exclude mortgage- and asset-backed securities, which are only reported from 1994.

One caveat on deposit rates is that U.S. bank regulations exist to restrict the rates that less-than-well-capitalized banks can offer to no more than 75 bps above the market rate (see Federal Deposit Insurance Act section 29 and FDIC rules and regulations section 325.103 for capital category definitions). These deposit rate caps are therefore expected to dampen the sensitivity of a bank's deposit rates to its riskiness. However, once a bank is under prompt corrective action, this effect should go against finding statistically and economically significant effects (as would a bank not wanting to signal the true extent of its adverse condition). It is nonetheless possible that a bank anticipating that it will soon be classified as less than well capitalized may have an incentive to be "active" earlier by raising its rates to attract deposits.

¹⁶ This cutoff follows the H8 criteria. Results are very similar if the cutoff is based on total deposits instead of total assets. Results are also similar for other cutoffs, such as the largest 10 banks.

C. The Relation between Deposit Rates and Liquidity Demand Risk

Table IV focuses on variation in a bank's vulnerability to liquidity demand shocks and its effect on deposit rates. The first two columns report the results for large time deposits followed by core deposits in columns (3) and (4). The results indicate that banks vulnerable to liquidity risk raised rates in the first year of the crisis. The key coefficient of interest is that on the interaction term of unused commitments with the dummy variable indicating the first year of the crisis, *Unused Commitment Ratio* $\times$ *Crisis1*. For example, banks with a greater amount of preexisting commitments outstanding offered significantly higher deposit rates in that period: columns (1) and (2) show that a 0.1 increase in the unused commitment ratio (roughly one standard deviation or the difference between a 75th percentile bank and a 25th percentile bank) raised the large time deposit rate by 5.9 to 7.1 bps. Banks exposed to liquidity demand even raised rates on core deposits in this phase of the crisis in a specification without bank fixed effects in column (4). As funding pressures eased in the second year of the crisis, these banks offered significantly lower rates in line with Gatev and Strahan (2006).

Note that the results do not support the alternative hypothesis that commitments-exposed banks are simply those with greater solvency problems. The results indicate that solvency problems, such as nonperforming loans and real estate loans (the latter are closely related to the proximate causes of the crisis), are independent risk factors. Indeed, their effects did not subside in the second year of the crisis, in contrast to the liquidity risk measures, further supporting the notion that unused commitments are properly picking up liquidity risk only. For example, a one-standard-deviation increase in a bank's real estate lending increased its core rate by 5.7 bps in the latter part of the crisis (column (3)). The difference in timing issues is parsed in greater detail in Section III.B.¹⁷ More generally, however, there can be an interesting interplay between liquidity risk and solvency risk, which we take up in Section III.

The remaining controls enter with the expected effects overall. For example, banks reliant on net wholesale funding increased deposit rates, also in the first year of the crisis when the liquidity shock was acute. Banks enjoying implicit support such as the TBTF institutions generally pay lower rates and were also associated with lower deposit rates on average during the crisis. For example, large banks lowered rates by 14 to 66 bps and the effect is statistically significant for core deposits (columns (3) and (4)).¹⁸

¹⁷ The Internet Appendix also shows results with a broad set of additional real estate and trading-related variables; the effect of unused commitments is robust. While a number of these measures are positively related to unused commitments, they are not strongly correlated. Moreover, none of these additional variables are associated with lending shortfalls so that their incentive to raise deposit rates was subdued or absent during the crisis.

¹⁸ Note also that the relation between commitments-exposure and deposit rates was positive for the set of larger banks, though weaker than for smaller banks (see the Internet Appendix). This result is in line with the generally muted sensitivity to liquidity risk by more "solvent" banks (see Section III). Note also that there is the possibility that a bank with a profitable business opportunity

Table IV
The Relationship between the Deposit Interest Rate and Liquidity Demand Risk in the Crisis

This table reports regressions relating the rate paid on deposits to a bank's liquidity demand risk during the crisis. The regressions allow for two phases of the crisis: 2007Q3 to 2008Q2 (*Crisis1*) and 2008Q3 to 2009Q2 (*Crisis2*). The dependent variable is the implicit rate on deposits, % annual (Call Reports). All specifications are panel regressions over the period 1994 to 2009 with fixed effects for bank organizations and quarterly time dummies (unless otherwise noted). The reported R^2 is within- R^2 . All regressions control for district time trends and for the deposit-weighted geographic market deposit concentration. The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions and details about the bank panel. ***, **, and * indicate 1%, 5%, and 10% significance, respectively.

	(1) Large Time	(2) Large Time	(3) Core	(4) Core
Unused Commitment Ratio _{<i>t</i>-1}	0.139 (0.095)	-0.810*** (0.106)	-0.329*** (0.093)	-1.804*** (0.117)
Unused Commitment Ratio _{<i>t</i>-1} × Crisis1	0.592*** (0.119)	0.710*** (0.139)	0.033 (0.095)	0.725*** (0.217)
Unused Commitment Ratio _{<i>t</i>-1} × Crisis2	-0.577*** (0.131)	-0.810*** (0.156)	-0.590*** (0.102)	-0.617*** (0.176)
Controls				
Net Wholesale Funding _{<i>t</i>-1}	0.310*** (0.043)	0.529*** (0.036)	0.012 (0.037)	0.890*** (0.044)
Net Wholesale Funding _{<i>t</i>-1} × Crisis1	0.119*** (0.046)	0.162*** (0.053)	0.467*** (0.042)	0.575*** (0.064)
Net Wholesale Funding _{<i>t</i>-1} × Crisis2	0.019 (0.051)	0.066 (0.058)	0.248*** (0.038)	0.312*** (0.059)
NPL to Loans _{<i>t</i>-1}	1.073*** (0.236)	0.687** (0.316)	0.218 (0.170)	1.122*** (0.346)
NPL to Loans _{<i>t</i>-1} × Crisis1	-0.549 (0.465)	0.302 (0.521)	0.784*** (0.269)	2.302** (0.952)
NPL to Loans _{<i>t</i>-1} × Crisis2	1.083*** (0.316)	1.872*** (0.395)	1.050*** (0.209)	2.373*** (0.457)
Capital Ratio _{<i>t</i>-1}	-0.834** (0.328)	-0.098 (0.223)	-1.834*** (0.260)	0.381 (0.259)
Capital Ratio _{<i>t</i>-1} × Crisis1	0.775*** (0.220)	0.660*** (0.233)	0.389 (0.306)	0.176 (0.417)
Capital Ratio _{<i>t</i>-1} × Crisis2	-0.418 (0.317)	-0.804** (0.354)	0.394* (0.218)	-0.542* (0.333)
Large Bank Indicator	-0.244* (0.146)	-0.204 (0.131)	-0.038 (0.122)	-0.393*** (0.087)
Large Bank Indicator × Crisis1	-0.167 (0.138)	-0.215 (0.163)	-0.307*** (0.073)	-0.664*** (0.093)
Large Bank Indicator × Crisis2	-0.142 (0.103)	-0.133 (0.143)	-0.002 (0.070)	-0.143 (0.094)
Real Estate Loan Share _{<i>t</i>-1}	0.025 (0.060)	0.117*** (0.042)	-0.044 (0.054)	-0.021 (0.050)
Real Estate Loan Share _{<i>t</i>-1} × Crisis1	-0.066 (0.053)	-0.069 (0.055)	0.058 (0.046)	0.003 (0.070)
Real Estate Loan Share _{<i>t</i>-1} × Crisis2	0.109* (0.062)	0.084 (0.077)	0.242*** (0.046)	0.067 (0.072)
Bank Fixed Effects	Yes	No	Yes	No
Observations	196,124	196,124	196,151	196,151
R^2	0.74	0.71	0.89	0.76

D. The Relation between Deposit Flows and Liquidity Demand Risk

We next turn to evaluating the responsiveness of deposit flows to a bank's liquidity risk, and whether the association changed in the crisis. The data on aggregate deposits suggest that the banking system as a whole was not particularly successful in attracting enough deposits in the first phase of the crisis. Such a shortfall in overall deposit inflows would be of special concern if deposit inflows at banks with high exposure to liquidity demand did not increase as much relative to other banks as in past crises (despite their offering higher rates as shown in the previous section).

The regression results confirm this to be the case in the first phase of the recent crisis, coinciding with the aggregate deposit funding squeeze relative to the increase in demand on banks for liquidity. Table V presents results of deposit growth regressions for several deposit types, including total deposits and measures of "stable" deposits (core deposits and insured deposits). For example, the interaction term of unused commitments with the first year of the crisis, *Unused Commitment Ratio* $\times$ *Crisis1*, enters negatively for both total and core deposit growth in columns (1) and (2) (implying that a 0.1 increase in the exposure to unused commitments was associated with a 0.16 to 0.17 percentage point decline in quarterly deposit and core deposit growth). This decline is economically significant when compared to an average quarterly deposit growth of 1.1 percentage points and core deposit growth of 0.8 percentage points (Table III). Therefore, banks with unused commitments lost their systematic advantage at gaining deposits, unlike during previous episodes of stress (as explored further in Section III). Even the inflow of insured deposits was small and statistically insignificant at the onset of the crisis (column (3)).¹⁹

Consistent with these findings, banks seeking brokered deposits and choosing to participate in the FDIC's TAG program at the end of 2008 were those that had a high unused commitment ratio (columns (4) and (5)). Breaking down the results by the type of deposit account indicates an active search for deposit funds by commitments-exposed banks, over and above that of a passive response to deposit withdrawals. For example, brokered deposits are funds obtained through a broker, whereby the broker often relies on a deposit listing service to identify high offer rates on deposits. The broker also often facilitates the placement of third-party deposits by selling participations in

will seek to raise deposit funding by offering a higher rate. This possibility, however, is unlikely to explain the results in this section—in particular, the shift in relative rates for riskier banks in the first year of the crisis. But this may be a plausible explanation for the positive estimated effect on the capital ratio (where the next section shows that well-capitalized banks increased lending growth initially more than deposit growth).

¹⁹ Note that the focus is on the interaction term of unused commitments with the crisis because this isolates a differences-in-differences effect, namely, whether there are significant deposit growth differences for banks before and after the crisis shock as a function of their treatment intensity (commitments exposure). The main conclusion in this paper is unaffected by instead examining the overall effect. Even if the reduced deposit growth was still overall positive (column (1)), the real issue is whether these commitments-exposed banks obtained sufficient deposit funding. The next section shows that this was not the case.

Table V
The Relationship between Deposit Inflows and Liquidity Demand Risk in the Crisis

This table reports regressions relating the growth of different categories of deposits to a bank's liquidity demand risk during the crisis. The regressions allow for two phases of the crisis: 2007Q3 to 2008Q2 (*Crisis1*) and 2008Q3 to 2009Q2 (*Crisis2*). Specifications (1) to (4) are panel regressions over the period 1994 to 2009 with fixed effects for bank organizations and quarterly time dummies. The reported R^2 is within- R^2 . The specification in column (5) is cross-sectional for 2008Q4 when the Transaction Account Guarantee (TAG) was introduced. Column (5) also controls for the share of transaction deposits. Regressions control for district time trends and for the deposit-weighted geographic market deposit concentration. The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions and details about the bank panel. ***, **, and * indicate 1%, 5%, and 10% significance, respectively.

	(1) $\Delta \text{Deposits}_t / \text{Assets}_{t-1}$	(2) $\Delta \text{Core Deposits}_t / \text{Assets}_{t-1}$	(3) $\Delta \text{Insured Deposits}_t / \text{Assets}_{t-1}$	(4) $\Delta \text{Brokered Deposits}_t / \text{Assets}_{t-1}$	(5) TAG Deposits (2008Q4) / Assets $_{t-1}$
Unused Commitment Ratio $_{t-1}$	0.034*** (0.004)	0.025*** (0.003)	0.014*** (0.002)	0.008*** (0.001)	0.174*** (0.016)
Unused Commitment Ratio $_{t-1} \times \text{Crisis1}$	-0.016*** (0.006)	-0.017*** (0.005)	0.007 (0.004)	0.007*** (0.002)	
Unused Commitment Ratio $_{t-1} \times \text{Crisis2}$	0.018*** (0.006)	0.005 (0.005)	0.030*** (0.004)	0.018*** (0.002)	
Controls					
Net Wholesale Funding $_{t-1}$	0.068*** (0.002)	0.066*** (0.002)	0.028*** (0.001)	0.008*** (0.000)	-0.023*** (0.007)
Net Wholesale Funding $_{t-1} \times \text{Crisis1}$	-0.008*** (0.002)	0.004** (0.002)	0.005*** (0.002)	0.002*** (0.001)	
Net Wholesale Funding $_{t-1} \times \text{Crisis2}$	0.005** (0.002)	0.010*** (0.002)	0.007*** (0.002)	0.006*** (0.001)	
NPL to Loans $_{t-1}$	-0.192*** (0.014)	-0.139*** (0.010)	-0.100*** (0.008)	-0.040*** (0.004)	0.073* (0.040)
NPL to Loans $_{t-1} \times \text{Crisis1}$	0.009 (0.015)	0.023 (0.015)	0.041*** (0.013)	0.014 (0.009)	
NPL to Loans $_{t-1} \times \text{Crisis2}$	0.058*** (0.019)	0.042*** (0.015)	0.056*** (0.013)	0.002 (0.006)	

(Continued)

Table V—Continued

	(1) $\Delta \text{Deposits}_t / \text{Assets}_{t-1}$	(2) $\Delta \text{Core Deposits}_t / \text{Assets}_{t-1}$	(3) $\Delta \text{Insured Deposits}_t / \text{Assets}_{t-1}$	(4) $\Delta \text{Brokered Deposits}_t / \text{Assets}_{t-1}$	(5) $\text{TAG Deposits (2008Q4)} / \text{Assets}_{t-1}$
Capital Ratio $_{t-1}$	0.189*** (0.012)	0.158*** (0.009)	0.058*** (0.007)	0.018*** (0.003)	−0.031 (0.029)
Capital Ratio $_{t-1} \times \text{Crisis1}$	0.024* (0.013)	0.023*** (0.011)	0.040*** (0.008)	0.007*** (0.003)	
Capital Ratio $_{t-1} \times \text{Crisis2}$	0.073*** (0.014)	0.053*** (0.012)	0.052*** (0.010)	0.022*** (0.004)	
Large Bank Indicator	−0.006** (0.003)	−0.005** (0.002)	−0.002 (0.002)	−0.002** (0.001)	0.013 (0.009)
Large Bank Indicator $\times$ Crisis1	0.005 (0.003)	0.000 (0.003)	0.000 (0.002)	0.000 (0.001)	
Large Bank Indicator $\times$ Crisis2	−0.005 (0.005)	0.009*** (0.003)	0.003 (0.002)	−0.005*** (0.001)	
Real Estate Loan Share $_{t-1}$	−0.012*** (0.002)	−0.012*** (0.002)	−0.006*** (0.001)	−0.001 (0.000)	−0.032*** (0.005)
Real Estate Loan Share $_{t-1} \times \text{Crisis1}$	−0.026*** (0.002)	−0.026*** (0.002)	−0.011*** (0.002)	0.002** (0.001)	
Real Estate Loan Share $_{t-1} \times \text{Crisis2}$	0.000 (0.003)	−0.004* (0.002)	0.003* (0.002)	0.002*** (0.001)	
Bank Fixed Effects	Yes	Yes	Yes	Yes	No
Observations	197,198	197,198	197,198	196,655	3,522
R ²	0.07	0.08	0.32	0.04	0.14

shares up to the deposit insurance limit in a given bank instrument to one or more investors. Similarly, the FDIC facilitated access to funds through its TAG program in which banks choosing to participate paid an extra premium for the additional insurance. For example, the estimated difference in the TAG share between the 75th and 25th percentile unused commitments-exposed bank (0.017) was close to half the actual difference between the 75th percentile and 25th percentile TAG share bank (0.04) (the results control for the share of transaction deposits).

Other results in Table V are mostly in line with priors: banks less reliant on wholesale funding gained deposits, as did better capitalized and large banks. Collectively, these findings imply that banks were not indiscriminately seen as safe havens during this crisis. As expected, insured deposit growth was less sensitive to measures of riskiness. For example, inflows into insured deposit accounts represented the bulk of inflows at banks with high nonperforming loans (column (3)).

E. The Relation between Lending, Shortfalls, and Liquidity Demand Risk

The final step is to test how liquidity-exposed banks adjust lending and other available margins on their balance sheet in reaction to the liquidity shock. The results in columns (1) and (2) of Table VI show that banks with preexisting commitments increased on-balance-sheet loan growth in the first year. This result reflects involuntary lending that occurs when off-balance-sheet commitments are converted to loans. The lending increase was not confined to increasing buybacks of conduit assets or to warehousing real estate loans but also reflected higher commercial and industrial (C&I) lending growth through loan commitment drawdowns by corporations (column (2)). For example, a high commitments-exposed bank saw a 0.14 percentage point greater increase in its quarterly loan growth than a low commitments-exposed bank (where high unused commitments are taken at the 75th percentile (0.17) and low commitments are taken at the 25th percentile (0.07)). A high commitments-exposed bank also saw a 0.04 percentage point increase in C&I loan growth (equivalent to 40% of the average quarterly increase in C&I loan growth).²⁰

As discussed earlier, Campello et al. (2011) documents compelling evidence of nonfinancial corporations' access to liquidity through extensive drawdowns of credit lines. Conducting a unique corporate survey during the crisis, they show that corporations drew heavily on credit lines and cancellations by banks were rare (even though banks increased commitment fees and shortened the maturity of the credit lines).²¹ Similarly, Berrospide, Meisenzahl, and Sullivan (2012) use SEC filings to show that nonfinancial firms were more likely to draw

²⁰ Additional evidence on C&I lending is available in the Internet Appendix, employing narrower measures of exposure to C&I commitments. The results also are robust to estimating over the set of nonconduit banks.

²¹ In an interesting parallel on the other side of the balance sheet, we find that the deposit maturity structure shortened for banks with higher liquidity demand risk. Also, banks that were tilted toward a short maturity profile before the crisis had greater rollover risk. They offered

Table VI

The Relationship between Lending, Loan-to-Deposit Shortfalls, and Liquidity Demand Risk in the Crisis

This table reports regressions relating lending growth and loan-to-deposit shortfalls to a bank's liquidity demand risk during the crisis. The regressions allow for two phases of the crisis: 2007Q3 to 2008Q2 (*Crisis1*) and 2008Q3 to 2009Q2 (*Crisis2*). All specifications are panel regressions over the period 1994 to 2009 with fixed effects for bank organizations and quarterly time dummies. The reported R^2 is within- R^2 . Regressions control for district time trends and for the deposit-weighted geographic market deposit concentration. The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions and details about the bank panel. ***, **, and * indicate 1%, 5%, and 10% significance, respectively.

	(1)	(2)	(3)	(4)	(5)
	$\Delta \text{ Loans}_t /$ Assets_{t-1}	$\Delta \text{ C\&I Loans}_t /$ Assets_{t-1}	$\Delta (\text{Loan} +$ $\text{Commitments})_t /$ $(\text{Assets} +$ $\text{Commitments})_{t-1}$	$(\text{Loans} -$ $\text{Deposits})_t /$ Assets_{t-1}	$\Delta (\text{Loans} -$ $\text{Deposits})_t /$ Assets_{t-1}
Unused Commitment Ratio $_{t-1}$	0.134*** (0.006)	0.018*** (0.001)	-0.016*** (0.004)	-0.042** (0.017)	0.099*** (0.006)
Unused Commitment Ratio $_{t-1} \times \text{Crisis1}$	0.014*** (0.005)	0.004** (0.002)	-0.023*** (0.005)	0.118*** (0.019)	0.028*** (0.007)
Unused Commitment Ratio $_{t-1} \times \text{Crisis2}$	-0.002 (0.006)	-0.003 (0.002)	-0.053*** (0.006)	0.095*** (0.017)	-0.021*** (0.007)
Controls					
Net Wholesale Funding $_{t-1}$	-0.012*** (0.001)	-0.003*** (0.000)	-0.020*** (0.001)	0.513*** (0.008)	-0.084*** (0.002)
Net Wholesale Funding $_{t-1} \times \text{Crisis1}$	-0.005*** (0.002)	-0.001** (0.001)	-0.010*** (0.002)	0.027*** (0.006)	0.003 (0.002)
Net Wholesale Funding $_{t-1} \times \text{Crisis2}$	-0.016*** (0.002)	-0.004*** (0.001)	-0.027*** (0.002)	0.014* (0.008)	-0.022*** (0.003)
NPL to Loans $_{t-1}$	-0.303*** (0.017)	-0.054*** (0.005)	-0.381*** (0.020)	-0.254*** (0.035)	-0.108*** (0.012)
NPL to Loans $_{t-1} \times \text{Crisis1}$	-0.069*** (0.017)	0.005 (0.007)	-0.087** (0.036)	-0.076 (0.051)	-0.077*** (0.018)
NPL to Loans $_{t-1} \times \text{Crisis2}$	0.018 (0.019)	0.018*** (0.005)	-0.014 (0.020)	-0.070* (0.039)	-0.046*** (0.016)

(Continued)

Table VI—Continued

	(1)	(2)	(3)	(4)	(5)
	$\Delta \text{Loans}_t / \text{Assets}_{t-1}$	$\Delta \text{C\&I Loans}_t / \text{Assets}_{t-1}$	$\Delta (\text{Loan} + \text{Commitments})_t / (\text{Assets} + \text{Commitments})_{t-1}$	$(\text{Loans} - \text{Deposits})_t / \text{Assets}_{t-1}$	$\Delta (\text{Loans} - \text{Deposits})_t / \text{Assets}_{t-1}$
Capital Ratio $_{t-1}$	0.015 (0.010)	0.004 (0.004)	0.022** (0.011)	0.725*** (0.053)	-0.176*** (0.014)
Capital Ratio $_{t-1} \times \text{Crisis1}$	0.042*** (0.010)	0.005 (0.004)	0.039*** (0.012)	0.045 (0.038)	0.027 (0.017)
Capital Ratio $_{t-1} \times \text{Crisis2}$	0.054*** (0.011)	0.010*** (0.004)	0.047*** (0.012)	0.010 (0.050)	-0.021 (0.017)
Large Bank Indicator	0.001 (0.003)	-0.001 (0.001)	-0.002 (0.003)	-0.006 (0.026)	0.007* (0.004)
Large Bank Indicator $\times \text{Crisis1}$	0.008*** (0.003)	0.005*** (0.001)	0.010*** (0.003)	-0.054*** (0.015)	0.003 (0.004)
Large Bank Indicator $\times \text{Crisis2}$	0.011*** (0.003)	0.005*** (0.001)	0.011*** (0.003)	-0.040*** (0.014)	0.018*** (0.005)
Real Estate Loan Share $_{t-1}$	0.004* (0.002)	0.011*** (0.001)	-0.003 (0.002)	0.039*** (0.010)	0.016*** (0.002)
Real Estate Loan Share $_{t-1} \times \text{Crisis1}$	0.001 (0.002)	0.003*** (0.001)	-0.007*** (0.002)	0.051*** (0.007)	0.028*** (0.003)
Real Estate Loan Share $_{t-1} \times \text{Crisis2}$	0.007*** (0.002)	0.009*** (0.001)	-0.004 (0.003)	0.037*** (0.008)	0.008*** (0.003)
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	197,198	197,198	197,198	197,198	197,198
R^2	0.15	0.03	0.11	0.50	0.10

down lines beginning in 2007 and to a much greater extent starting in March 2008 following Bear Stearns's problems, while credit line cancellations and covenant-induced reductions were small during the crisis. Santos (2012) uses confidential data on large corporations' syndicated credit lines to also show that higher drawdowns occurred during the 2007 to 2009 crisis, especially on riskier banks.

However, liquidity-exposed banks significantly reduced overall credit (the sum of loans and commitments) as they cut back on the extension of new loan commitments and new term loans (column (3) of Table VI). For example, a 0.1 increase in a bank's unused commitment ratio led to a 0.23 percentage point decrease in new credit growth compared with a typically positive credit growth of 1.1 percentage points (Table III). Similar evidence of drawdowns displacing new credit originations is documented by Cornett et al. (2011), Ivashina and Scharfstein (2010), and Mora (2010).

Importantly, while commitments-exposed banks managed to fund drawn credit lines, the shortfall between on-balance-sheet lending and deposits *widened* in the first year as shown in columns (4) and (5) of Table VI. The shortfall is expressed here as the difference between loans and deposits scaled by lagged assets in column (4) and as the growth rate of this term in column (5). For example, a 0.1 increase in undrawn commitments led to a 0.28 percentage point increase in the quarterly difference between lending and deposit growth (column (5)), compared to a typical zero value. These results show that the synergy broke down in the first year, and thus banks were unable to meet their lending needs with deposit funding alone and were forced to turn to other margins of adjustment. This shortfall subsided with the aggregate liquidity reversal in the banking system as a whole during the second year of the crisis. In contrast, the shortfall persisted at real-estate-exposed banks through the two periods of the crisis.

To analyze how banks met the demands on their liquidity-providing services, in Table VII we show the possible margins of adjustment available to liquidity-short banks. These include running down liquid asset buffers, borrowing from the interbank and repo markets, and borrowing from other sources known as "other borrowed money" in the Call Reports—primarily government and government-sponsored agencies. First, the result in column (1) shows that commitments-exposed banks drew on their liquid asset buffers as reflected by the negative coefficient on the interaction term *Unused Commitment Ratio* $\times$ *Crisis1*. Second, the key result in column (3) is that a high commitments-exposed bank significantly increased its growth of other borrowed money by 0.13 percentage points more than a low commitments-exposed bank in the first year, primarily from the FHLBs (a 0.11 percentage point increase as seen in column (4)). To make sense of the relative importance of this government-related borrowing, one can compare it to interbank borrowing. The results in columns (2) and (3) show that reintermediation within the banking system

significantly higher rates in the first year of the crisis and also sought to lengthen the maturity of their time deposits during the crisis (see the Internet Appendix).

Table VII

The Relationship between Liquid Assets, Borrowed Money, and Liquidity Demand Risk in the Crisis

This table reports regressions relating the growth of liquid assets and nondeposit borrowing to a bank's liquidity demand risk during the crisis. The regressions allow for two phases of the crisis: 2007Q3 to 2008Q2 (*Crisis1*) and 2008Q3 to 2009Q2 (*Crisis2*). The dependent variable in column (3), "other borrowed money" (RCFD3190), is the sum of borrowing from the FHLBs (column (4)) and "other" (includes borrowing from the Federal Reserve) (column (5)), which are available from 2001 only (see Call Reports schedule RC-M). All specifications are panel regressions over the period 1994 to 2009 (2001 to 2009 in columns (4) and (5)) with fixed effects for bank organizations and quarterly time dummies. The reported R^2 is within- R^2 . Regressions control for district time trends and for the deposit-weighted geographic market deposit concentration. The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions and details about the bank panel. ***, **, and * indicate 1%, 5%, and 10% significance, respectively.

	(1)	(2)	(3)	(4)	(5)
	Δ (Liquid Assets) _{<i>t</i>} / Assets _{<i>t</i>-1}	Δ (Federal Funds and Repo Borrowing) _{<i>t</i>} / Assets _{<i>t</i>-1}	Δ (Other Borrowed Money Total) _{<i>t</i>} / Assets _{<i>t</i>-1}	Δ (Other Borrowed Money FHLB) _{<i>t</i>} / Assets _{<i>t</i>-1}	Δ (Other Borrowed Money Other) _{<i>t</i>} / Assets _{<i>t</i>-1}
Unused Commitment Ratio _{<i>t</i>-1}	-0.073*** (0.005)	0.007*** (0.001)	0.012*** (0.002)	0.016*** (0.002)	0.001 (0.000)
Unused Commitment Ratio _{<i>t</i>-1} × Crisis1	-0.011* (0.006)	0.004*** (0.002)	0.013*** (0.002)	0.011*** (0.002)	0.001* (0.001)
Unused Commitment Ratio _{<i>t</i>-1} × Crisis2	0.009 (0.006)	-0.009*** (0.002)	-0.003 (0.003)	-0.004* (0.002)	0.002** (0.001)
Controls					
Net Wholesale Funding _{<i>t</i>-1}	0.067*** (0.002)	-0.010*** (0.000)	-0.010*** (0.001)	-0.013*** (0.001)	-0.001*** (0.000)
Net Wholesale Funding _{<i>t</i>-1} × Crisis1	0.002 (0.002)	0.000 (0.001)	0.005*** (0.001)	0.004*** (0.001)	0.001** (0.000)
Net Wholesale Funding _{<i>t</i>-1} × Crisis2	0.008*** (0.002)	-0.003*** (0.001)	-0.011*** (0.001)	-0.010*** (0.001)	0.000 (0.000)
NPL to Loans _{<i>t</i>-1}	0.009 (0.010)	-0.004** (0.002)	-0.019*** (0.003)	-0.013*** (0.004)	-0.001 (0.001)
NPL to Loans _{<i>t</i>-1} × Crisis1	0.034* (0.018)	-0.012*** (0.004)	-0.020*** (0.008)	-0.024*** (0.009)	-0.001 (0.001)
NPL to Loans _{<i>t</i>-1} × Crisis2	0.007 (0.016)	0.000 (0.003)	0.009* (0.005)	0.007 (0.005)	0.000 (0.001)

(Continued)

Table VII—Continued

	(1)	(2)	(3)	(4)	(5)
	Δ (Liquid Assets) _t / Assets _{t-1}	Δ (Federal Funds and Repo Borrowing) _t / Assets _{t-1}	Δ (Other Borrowed Money Total) _t / Assets _{t-1}	Δ (Other Borrowed Money FHLB) _t / Assets _{t-1}	Δ (Other Borrowed Money Other) _t / Assets _{t-1}
Capital Ratio _{t-1}	0.177*** (0.013)	0.006*** (0.002)	0.020*** (0.004)	0.024*** (0.004)	0.001 (0.001)
Capital Ratio _{t-1} × Crisis1	-0.021 (0.014)	-0.001 (0.002)	0.007 (0.007)	0.001 (0.004)	0.003 (0.002)
Capital Ratio _{t-1} × Crisis2	0.028* (0.015)	0.005* (0.003)	0.007 (0.005)	0.005 (0.004)	0.003* (0.002)
Large Bank Indicator	-0.007*** (0.003)	-0.001 (0.001)	0.000 (0.001)	0.002 (0.002)	-0.001 (0.001)
Large Bank Indicator × Crisis1	0.001 (0.003)	0.000 (0.001)	0.003 (0.002)	0.001 (0.002)	0.003** (0.001)
Large Bank Indicator × Crisis2	-0.016*** (0.005)	0.000 (0.002)	0.000 (0.001)	0.005*** (0.001)	-0.002 (0.001)
Real Estate Loan Share _{t-1}	-0.012*** (0.002)	0.001*** (0.000)	0.002** (0.001)	0.001 (0.001)	0.0003 (0.002)
Real Estate Loan Share _{t-1} × Crisis1	-0.022*** (0.002)	0.001* (0.001)	0.005*** (0.001)	0.006*** (0.001)	0.0003 (0.002)
Real Estate Loan Share _{t-1} × Crisis2	-0.007*** (0.002)	0.002*** (0.001)	0.001 (0.001)	0.001 (0.001)	0.0003 (0.003)
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	197,198	197,198	197,198	122,324	122,324
R ²	0.07	0.02	0.03	0.03	0.01

from liquidity-rich to liquidity-poor banks through the interbank and repo markets comprised only one-quarter of their total nondeposit borrowing growth (i.e., 0.04 of a 0.17 percentage point total).²² Or, more simply put, almost the entire increase in on-balance-sheet lending growth for liquidity-exposed banks (0.14 percentage points from column (1) of Table VI) was funded with explicit government borrowing.

III. Additional Results and Robustness Tests

A. Do Liquidity and Solvency Problems Interact?

Above we show empirically that a bank's exposure to liquidity demand risk is an important determinant of its deposit funding position, its rates on deposits, and the extent to which the bank takes offsetting draws on its liquid assets and borrowing actions, controlling for a range of solvency risk measures. Therefore, at a basic level, the hypothesis that a bank's liquidity risk can be collapsed into solvency risk is rejected. However, theory predicts a relation between liquidity risk and solvency risk, even when these two risks can be easily separated conceptually. For example, in the model by Rochet and Vives (2004), an otherwise solvent bank may be illiquid when investors receive a negative signal on the future realization of the return on the bank's assets and withdraw early. In their model, there is an intermediate range of fundamentals (solvency) over which a coordination failure can occur. But this range tightens as the strength of fundamentals increases or as the adverse selection premium on a bank's asset sales declines. Similarly, in the model by Diamond and Rajan (2005), a higher risk of insolvency leads to a socially detrimental outcome because the bank is forced to sell not only the late project loans but also other loans that would soon deliver real liquidity and for which the banker has an advantage in collection skills, exacerbating an aggregate liquidity shortage.

Therefore, a testable hypothesis that flows directly from the theory is that a bank's vulnerability to liquidity risk is expected to be greater within the class of banks with greater solvency problems. That is, the estimated coefficient on *Unused Commitment Ratio* $\times$ *Crisis1* should be greater in magnitude for the class of banks characterized by weak fundamentals. An alternative hypothesis is that liquidity risk has a uniform effect irrespective of the bank's solvency position. This latter hypothesis is consistent with the view that the *Crisis1* funding shock reflected a concern about the asset quality, and therefore insolvency, of the banking system as a whole.

²² That is, the 0.28 percentage point increase in the lending growth shortfall (from Table VI) was covered by a 0.13 percentage point increase in other borrowing, a 0.11 percentage point decrease in liquid assets, and a 0.04 percentage point increase in interbank borrowing. At the same time, there is evidence that commitments-exposed banks significantly withdrew from lending on the interbank market until 2008Q3. Afonso, Kovner, and Schoar (2011) also show that the interbank market rationed riskier banks and did not expand to meet latent demand. Note that banks with much real estate lending covered most of their shortfall by running down liquid assets (including into the second year when there was no inflow of deposit funds at real estate banks).

Table VIII presents the results of these tests in which the sample is partitioned into high- and low-solvency problems according to three proxies for fundamentals: high nonperforming loans, high real estate loans, and low capital. The top panel shows the rates on large time deposits and the lower panel shows deposit funding flows. The results generally support the hypothesis that weaker banks react to the aggregate funding shock by raising deposit rates more than fundamentally stronger banks with a similar *ex ante* liquidity demand exposure. For example, a 0.1 increase in unused commitments raised the rate of banks with high nonperforming loans (real estate loans) by 6.9 (6.5) bps compared with 4.6 (3.1) bps for banks with low nonperforming loans (real estate loans). In the case of real estate loans, these effects are found to be statistically significantly different as well, so that a uniform effect of liquidity demand risk is rejected. To summarize, bank liquidity risk can interact with solvency risk in practice.

A related question is whether we observe that banks' advantage as liquidity providers is restored in this crisis when focusing on the set of fundamentally stronger banks. It is plausible that these banks were still seen as a safe haven and able to reintermediate excess funds (Gatev and Strahan (2006)). Interestingly, however, the deposit growth regressions in Panel B of Table VIII show no evidence of a compensating effect within the class of fundamentally stronger banks. In other regressions (see the Internet Appendix), we also confirm that these stronger but liquidity-exposed banks did not significantly increase interbank and repo lending to other banks in the crisis.²³

B. Robustness Checks

This section offers additional support for our thesis that the liquidity-insurance mechanism broke down at commitments-exposed banks precisely when there was a shortage of funding in the banking system. Moreover, the liquidity shock constrained banks that experienced higher *actual* drawdowns during the crisis.

B.1. Timing Issues

The empirical model in Table IX exploits the aggregate deposit funding shift after Lehman's failure by deconstructing the *Crisis2* dummy into the 2008Q3 quarter and the remainder of the *Crisis2* period. One alternative is that the observed easing in *Crisis2* could be spuriously picking up something that occurred later in the crisis, not attributable to the turnaround in liquidity following Lehman's failure. The results show that, indeed, the very sharp increase in deposits (and liquid assets) at liquidity-exposed banks occurred in 2008Q3, not later in the period (when comparing the coefficients on *Unused Commitment Ratio* $\times$ *Crisis2,08Q3* and *Unused Commitment Ratio* $\times$ *Crisis2,08Q4-09Q2* in

²³ There is some evidence, however, that stronger but liquidity-exposed banks were better able to borrow on the interbank market to cover their loan-to-deposit shortfalls than their fundamentally weaker peers (although they still obtained the majority of their nondeposit funding from other borrowed money). The fundamentally weaker banks resorted more to running down liquid assets and borrowing from other sources.

Table VIII
The Relationship between Solvency Risk and Liquidity Demand Risk in the Crisis: Are Banks with Solvency Problems More Vulnerable to Liquidity Demand Risk?

This table reports regressions testing whether banks with weaker fundamentals were more vulnerable to the onset of the crisis. The regressions allow for two phases of the crisis: 2007Q3 to 2008Q2 (*Crisis1*) and 2008Q3 to 2009Q2 (*Crisis2*). The sample is partitioned into high- and low-solvency problems based on nonperforming loans (columns (1) and (2)), real estate loans (columns (3) and (4)), or capital (columns (5) and (6)). “High” solvency risk is proxied by the set of banks with an above-median nonperforming loan ratio, above-median real estate loan share, or below-median capital ratio (Table III). All specifications are panel regressions over the period 1994 to 2009 with fixed effects for bank organizations and quarterly time dummies. The reported R^2 is within- R^2 . Regressions control for district time trends and for the deposit-weighted geographic market deposit concentration. The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions and details about the bank panel. ***, **, and * indicate 1%, 5%, and 10% significance, respectively.

Panel A: Rate on Large Time Deposits						
	(1) Nonperforming Loans High	(2) Nonperforming Loans Low	(3) Real Estate Loan Share High	(4) Real Estate Loan Share Low	(5) Capital Ratio High	(6) Capital Ratio Low
Unused Commitment Ratio _{<i>t</i>-1}	0.049 (0.129)	0.368*** (0.127)	0.210* (0.114)	0.087 (0.150)	0.173 (0.148)	0.043 (0.124)
Unused Commitment Ratio _{<i>t</i>-1} × Crisis1	0.688*** (0.159)	0.456*** (0.170)	0.651*** (0.140)	0.306* (0.184)	0.577*** (0.191)	0.530*** (0.147)
Unused Commitment Ratio _{<i>t</i>-1} × Crisis2	-0.351** (0.152)	-0.615*** (0.222)	-0.586*** (0.158)	-0.631*** (0.212)	-0.153 (0.209)	-0.826*** (0.159)
Other bank controls included	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	99,731	96,393	111,204	84,920	84,466	111,658
R ²	0.72	0.74	0.72	0.72	0.71	0.73

(Continued)

Table VIII—Continued

	Panel B. Total Deposit Growth					
	(7) Nonperforming Loans High	(8) Nonperforming Loans Low	(9) Real Estate Loan Share High	(10) Real Estate Loan Share Low	(11) Capital Ratio High	(12) Capital Ratio Low
Unused Commitment Ratio _{<i>t</i>-1}	0.029*** (0.005)	0.037*** (0.005)	0.063*** (0.005)	0.009* (0.005)	0.036*** (0.005)	0.028*** (0.005)
Unused Commitment Ratio _{<i>t</i>-1} × Crisis1	-0.016** (0.007)	-0.013 (0.009)	-0.029*** (0.006)	-0.001 (0.010)	-0.015 (0.009)	-0.015** (0.008)
Unused Commitment Ratio _{<i>t</i>-1} × Crisis2	0.019** (0.008)	0.011 (0.010)	0.020*** (0.007)	0.017* (0.009)	0.021* (0.011)	0.022*** (0.007)
Other bank controls included	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	100,013	97,185	111,329	85,869	85,178	112,020
R ²	0.08	0.07	0.08	0.08	0.09	0.07

Table IX
The Aggregate Shift in Funding: Separating Crisis2 into 2008Q3 (Post-Lehman) and 2008Q4 to 2009Q2

This table reports regressions relating the rate paid on deposits, changes in funding, and lending to a bank's liquidity demand risk during the crisis. The regressions allow for three phases of the crisis: 2007Q3 to 2008Q2 (*Crisis1*), 2008Q3 (*Crisis2 08Q3*), and 2008Q4 to 2009Q2 (*Crisis2 08Q4-09Q2*). All specifications are panel regressions over the period 1994 to 2009 with fixed effects for bank organizations and quarterly time dummies. The reported R^2 is within- R^2 . Regressions control for district time trends and for the deposit-weighted geographic market deposit concentration. The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions, details about the bank panel, and for full controls. ***, **, and * indicate 1% and 10% significance, respectively

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Rate on Large Time Deposits	Rate on Core Deposits	Δ Deposits _{<i>t</i>} / Assets _{<i>t-1</i>}	Δ (Loans – Deposits) _{<i>t</i>} / Assets _{<i>t-1</i>}	Δ (Liquid Assets) _{<i>t</i>} / Assets _{<i>t-1</i>}	Δ (Federal Funds and Repo Borrowing) _{<i>t</i>} / Assets _{<i>t-1</i>}	Δ (Other Borrowed Money Total) _{<i>t</i>} / Assets _{<i>t-1</i>}
Unused Commitment Ratio _{<i>t-1</i>}	0.140 (0.095)	-0.329*** (0.093)	0.034*** (0.004)	0.099*** (0.006)	-0.073*** (0.005)	0.008*** (0.001)	0.012*** (0.002)
Unused Commitment Ratio _{<i>t-1</i>} × Crisis1	0.594*** (0.120)	0.033 (0.095)	-0.016*** (0.006)	0.028*** (0.007)	-0.011* (0.006)	0.004*** (0.002)	0.013*** (0.002)
Unused Commitment Ratio _{<i>t-1</i>} × Crisis2 08Q3	-0.315* (0.174)	-0.541*** (0.109)	0.049*** (0.010)	-0.058*** (0.012)	0.030*** (0.010)	-0.015*** (0.004)	-0.007 (0.005)
Unused Commitment Ratio _{<i>t-1</i>} × Crisis2 08Q4-09Q2	-0.656*** (0.145)	-0.605*** (0.112)	0.008 (0.007)	-0.007 (0.008)	0.002 (0.007)	-0.007*** (0.002)	-0.002 (0.003)
Other bank controls included	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	196,124	196,151	197,198	197,198	197,198	197,198	197,198
R^2	0.74	0.89	0.08	0.10	0.07	0.02	0.03

columns (3) and (5)). For example, a 0.1 increase in a bank's exposure to unused commitments was associated with a 0.49 percentage point increase in deposit growth for the period ending September 2008 but only a 0.08 percentage point increase in the subsequent three quarters, on average.

The crisis indicators can also be further partitioned into their respective four quarters (results are shown in the Internet Appendix in the interest of space). The results confirm that the largest deposit inflow at liquidity-exposed banks occurred in 2008Q3, and was sustained by a smaller, significant increase in 2008Q4. At the same time, the lending shortfall significantly reversed in 2008Q3 after having peaked in 2008Q1. What helped close the loan-to-deposit gap was sustained growth of other borrowed money throughout the 2007Q4 to 2008Q2 period. In addition to the reintermediation of funds by the FHLBs, the Federal Reserve had explicitly put in place the Term Auction Facility (TAF) and other extensive term funding programs such as the Term Securities Lending Facility (TSLF) by mid-2008.²⁴ For example, the TAF provided credit to institutions with access to the discount window but at an auction-determined rate to mitigate stigma.

To summarize, these results support the view that explicit government intervention propped up liquidity-exposed banks. In this light, the funding outflows from prime money funds following Lehman's failure can be seen as a supporting factor that enforced the funding availability at commitments-exposed banks and helped lower their deposit rates further. The results also are robust to the use of precrisis values for commitments and other controls, mitigating the concern that these measures are contaminated by regulatory changes and bank-specific actions taken after the onset of the crisis to change risk profiles (see the Internet Appendix).

An alternative data source, which is based on a proprietary survey of current rates (mostly on CDs) of banks and thrifts conducted at the weekly frequency by Bank Rate Monitor® (BRM) can also be used to study rate changes in a window surrounding the Lehman event. This survey has a number of advantages and disadvantages. Its main advantage is that reported rates reflect *marginal* funding costs rather than *average* funding costs as measured using the Call Reports data. However, the survey corresponds to a selection of banks and is largely participation based. The interesting result (see the Internet Appendix) is that, unlike solvency measures (such as nonperforming and real estate loans), liquidity risk was highest in the period just before Lehman failed. Rates eased for commitments-exposed banks in the aftermath and then more so after TARP was introduced.²⁵

²⁴ As noted by Ashcraft, Bech, and Frame (2010), by end-2007 it became evident that the FHLB system was not enough to ease liquidity stress, so that by May 2008 the Federal Reserve became the largest government-sponsored liquidity facility. It is worth noting, therefore, that these programs helped support commitments-exposed banks, thus easing their pursuit of customer deposits by mid-2008. For example, commitments-exposed banks raised deposit rates (even core rates) at the onset of the crisis, but this pressure appears to have subsided by 2008Q2.

²⁵ This is not to say that nonperforming loans did not push up bank CD rates throughout the crisis, just that the relation did not die out suddenly in the aftermath of Lehman's failure. See

B.2. Effective Drawdowns

We next hypothesize that, if unused commitments capture liquidity risk, the liquidity shock should have constrained those banks that experienced significant drawdowns on credit lines during the crisis, not just any bank with a lot of credit lines outstanding (i.e., the relevant measure is effective, not potential, drawdowns). It is this constrained set that should have relied most on other borrowing and running down liquid buffers as well as raising deposit rates in an environment of tight funding. This is precisely what we find in Table X (full results for the controls are in the Internet Appendix) by comparing the coefficients on *Unused Commitment Ratio* $\times$ *Crisis1* for banks with above- and below-median declines in their unused commitments during the crisis. For instance, columns (11) and (12) show that the set of banks with larger effective drawdowns cut their liquid asset growth by 0.27 percentage points for a 0.1 increase in ex ante exposure to commitments, while the set with smaller drawdowns did not significantly adjust their liquidity margins. At the same time, because banks honored their commitments, the on-balance-sheet lending of the constrained set of banks significantly increased as expected despite all indications that they were more stressed, meaning that this result is hard to reconcile with a voluntary expansion in lending by this group (columns (5) and (6)).

The results also show that, while commitments-exposed banks lowered rates in the latter part of the crisis when funding was aplenty (a roughly equal decline for banks with above- and below-median effective drawdowns), the above-median set saw the sharpest increase in rates at the onset (coefficients of 0.75 vs. 0.39 in columns (1) and (2)).

C. Previous Crises: Revisiting Gatev and Strahan (2006)

We also extend the Gatev and Strahan (2006) results, confirming the hedge against market dislocations that banks provided in previous crises such as when LTCM failed but further showing how deposits growth and real-sector credit were disrupted when commercial paper spreads rose in this crisis. To ensure that differences in model specification are not driving the results, the specifications in Tables XI and XII closely follow those reported in Gatev and Strahan (2006). The key coefficient of interest is the interaction term of unused commitments with the commercial paper spread, which proxies for financial market stress. The coefficients on *Unused Commitments* $\times$ *Stress* are positive and statistically significant in both the deposit growth (Table XI, column (1)) and loan growth (Table XII, columns (1) and (4)) equations for a sample from 1991 to 2000 as in Gatev and Strahan.

We extend the model by allowing for a different effect stemming from a bank's unused commitments on its deposits and loans in this crisis compared with previous episodes of high commercial paper spreads, and thus the sample period is extended to 2009. The change in the influence of financial stress on

the Internet Appendix for results over a wider period, comparable to the relation between deposit rates and bank fundamentals in Table IV from Call Report data.

Table X
Partitioning Banks by Their Unused Commitments Constraint: Effective Drawdowns

This table reports regressions testing the hypothesis that the crisis should have especially constrained banks with greater effective, not potential, drawdowns. The regressions allow for two phases of the crisis: 2007Q3 to 2008Q2 (*Crisis1*) and 2008Q3 to 2009Q2 (*Crisis2*). Note that “High” regressions is proxied by the set of banks with more than four quarters during the eight-quarter crisis of negative growth of their available commitments. These banks also roughly correspond to banks with above-median declines in unused commitments. All specifications are panel regressions over the period 1994 to 2009 with fixed effects for bank organizations and quarterly time dummies. The reported R^2 is within- R^2 . Regressions control for district time trends and for the deposit-weighted geographic market deposit concentration. The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions, details about the bank panel, and the full details for the controls. *** and ** indicate 1% and 5% significance, respectively.

	(1) Rate on Large Time Deposits High	(2) Rate on Large Time Deposits Low	(3) Δ Deposits _{<i>t</i>} / Assets _{<i>t</i>-1} High	(4) Δ Deposits _{<i>t</i>} / Assets _{<i>t</i>-1} Low	(5) Δ Loans _{<i>t</i>} / Assets _{<i>t</i>-1} High	(6) Δ Loans _{<i>t</i>} / Assets _{<i>t</i>-1} Low
Unused Commitment Ratio _{<i>t</i>-1}	0.133 (0.141)	0.090 (0.148)	0.048*** (0.006)	0.022*** (0.005)	0.146*** (0.011)	0.113*** (0.009)
Unused Commitment Ratio _{<i>t</i>-1} × Crisis1	0.748*** (0.160)	0.390** (0.175)	-0.021*** (0.007)	-0.009 (0.009)	0.030*** (0.007)	0.008 (0.006)
Unused Commitment Ratio _{<i>t</i>-1} × Crisis2	-0.555*** (0.212)	-0.619*** (0.164)	0.015 (0.010)	0.019*** (0.007)	0.007 (0.008)	-0.007 (0.008)
Other bank controls included	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	68,511	83,442	68,645	84,105	68,645	84,105
R ²	0.78	0.75	0.09	0.07	0.22	0.12

(Continued)

Table X—Continued

	(7)	(8)	(9)	(10)	(11)	(12)
	Δ Brokered Deposits/ Assets _{<i>t</i>-1} High	Δ Brokered Deposits/ Assets _{<i>t</i>-1} Low	Δ (Other Borrowed Money Total) _{<i>t</i>} / Assets _{<i>t</i>-1} High	Δ (Other Borrowed Money Total) _{<i>t</i>} / Assets _{<i>t</i>-1} Low	Δ (Liquid Assets) _{<i>t</i>} / Assets _{<i>t</i>-1} High	Δ (Liquid Assets) _{<i>t</i>} / Assets _{<i>t</i>-1} Low
Unused Commitment Ratio _{<i>t</i>-1}	0.011*** (0.002)	0.006*** (0.001)	0.011*** (0.002)	0.011*** (0.003)	-0.073*** (0.008)	-0.068*** (0.007)
Unused Commitment Ratio _{<i>t</i>-1} $\times$ Crisis1	0.010*** (0.003)	0.004 (0.004)	0.017*** (0.004)	0.010*** (0.003)	-0.027*** (0.006)	-0.001 (0.009)
Unused Commitment Ratio _{<i>t</i>-1} $\times$ Crisis2	0.026*** (0.003)	0.011*** (0.003)	-0.001 (0.004)	-0.003 (0.003)	-0.001 (0.009)	0.016** (0.008)
Other bank controls included	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	68,567	83,787	68,645	84,105	68,645	84,105
R ²	0.06	0.03	0.04	0.03	0.07	0.08

Table XI

The Relationship between Market Stress and Growth of Deposits and Nondeposit Liabilities: Was It Different in the 2007 to 2009 Crisis? Revisiting Gatev and Strahan (2006)

This table reports regressions relating the growth of deposit and nondeposit liabilities to a bank's liquidity demand risk during market stress episodes. *Stress* is proxied by the commercial paper spread, *Crisis1* is a dummy variable equal to one from 2007Q3 to 2008Q2, and *Crisis2* is a dummy variable equal to one from 2008Q3 to 2009Q2. All specifications are panel regressions with fixed effects for bank organizations and quarterly time dummies. The sample period is from 1990 to 2009, except for the Gatev and Strahan sample (1991 to 2000). Following Gatev and Strahan, the interaction of the three-month Treasury bill rate with the unused commitments ratio is also controlled for. The reported R^2 is within- R^2 . The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions and details about the bank panel. ***, **, and * indicate 1%, 5%, and 10% significance, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Δ Deposits/ Assets_{t-1}		Δ Nondeposit Liabilities/ Assets_{t-1}			
	Gatev and Strahan sample 1991 to 2000	1990 to 2009	1990 to 2009	Gatev and Strahan sample 1991 to 2000	1990 to 2009	1990 to 2009
Unused Commitment Ratio_{t-1}	-0.021*** (0.007)	0.025*** (0.003)	0.025*** (0.003)	0.021*** (0.005)	0.013*** (0.002)	0.016*** (0.002)
Unused Commitment $\text{Ratio}_{t-1} \times \text{Stress}$	0.041*** (0.011)	0.007 (0.005)	0.022*** (0.008)	-0.004 (0.008)	0.006* (0.003)	-0.007 (0.006)
Unused Commitment $\text{Ratio}_{t-1} \times \text{Stress} \times \text{Crisis1}$			-0.039*** (0.017)			0.029*** (0.013)
Unused Commitment $\text{Ratio}_{t-1} \times \text{Stress} \times \text{Crisis2}$			0.018 (0.013)			0.014 (0.009)
Unused Commitment $\text{Ratio}_{t-1} \times \text{Crisis1}$			0.004 (0.014)			-0.003 (0.009)
Unused Commitment $\text{Ratio}_{t-1} \times \text{Crisis2}$			-0.011 (0.007)			-0.016*** (0.004)
Controls						
Capital ratio and interactions with stress and crisis	Yes	Yes	Yes	Yes	Yes	Yes
Size and interactions with stress and crisis	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	114,323	252,526	252,526	114,323	252,526	252,526
R^2	0.05	0.04	0.04	0.02	0.03	0.03

Table XII
The Relationship between Market Stress and Growth of Loans and Total Credit: Was It Different in the 2007 to 2009 Crisis? Revisiting Gatev and Strahan (2006)

This table reports regressions relating the growth of loans and total credit (loans plus commitments) to a bank's liquidity demand risk during market stress episodes. *Stress* is proxied by the commercial paper spread, *Crisis1* is a dummy variable equal to one from 2007Q3 to 2008Q2, and *Crisis2* is a dummy variable equal to one from 2008Q3 to 2009Q2. All specifications are panel regressions with fixed effects for bank organizations and quarterly time dummies. The sample period is from 1990 to 2009, except for the Gatev and Strahan sample (1991 to 2000). Following Gatev and Strahan, the interaction of the three-month Treasury bill rate with the unused commitments ratio is also controlled for. The reported R^2 is within- R^2 . The standard errors used in calculating significance levels are clustered at the bank organization level and are shown in parentheses. See the Internet Appendix for variable definitions and details about the bank panel. *** and * indicate 1% and 10% significance, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	$\Delta \text{ Loans}_t / \text{Assets}_{t-1}$			$\Delta (\text{Loan} + \text{Commitments})_t / (\text{Assets} + \text{Commitments})_{t-1}$		
	Gatev and Strahan sample 1991 to 2000	1990 to 2009	1990 to 2009	Gatev and Strahan sample 1991 to 2000	1990 to 2009	1990 to 2009
Unused Commitment Ratio $_{t-1}$	0.087*** (0.010)	0.110*** (0.007)	0.113*** (0.007)	-0.009 (0.010)	0.014*** (0.004)	0.023*** (0.004)
Unused Commitment Ratio $_{t-1} \times \text{Stress}$	0.047*** (0.012)	0.004 (0.005)	0.032*** (0.009)	0.031*** (0.011)	-0.035*** (0.006)	0.018* (0.009)
Unused Commitment Ratio $_{t-1} \times \text{Stress} \times \text{Crisis1}$			-0.020 (0.018)			0.026 (0.018)
Unused Commitment Ratio $_{t-1} \times \text{Stress} \times \text{Crisis2}$			-0.023* (0.013)			-0.026* (0.015)
Unused Commitment Ratio $_{t-1} \times \text{Crisis1}$			0.013 (0.014)			-0.067*** (0.014)
Unused Commitment Ratio $_{t-1} \times \text{Crisis2}$			-0.004 (0.006)			-0.053*** (0.008)
Controls						
Capital ratio and interactions with stress and crisis	Yes	Yes	Yes	Yes	Yes	Yes
Size and interactions with stress and crisis	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	114,323	252,526	252,526	114,323	252,526	252,526
R^2	0.05	0.08	0.08	0.02	0.05	0.05

banks' ability to provide liquidity in the recent crisis is captured by the triple interaction *Unused Commitments* $\times$ *Stress* $\times$ *Crisis*. The results in columns (3) of Tables XI and XII show that the coefficients on *Unused Commitments* $\times$ *Stress* $\times$ *Crisis* are mostly negative and statistically significant.²⁶ Together, the deposit and credit results suggest that the very banks that had offered insurance to businesses and households before the crisis in the form of commitments were not as well positioned to deliver the promised liquidity. And, while nondeposit funding played no regular role in supporting the banking system in previous crises (see also Gatev and Strahan (2006)), this occurred *throughout* the recent crisis (columns (4) to (6) of Table XI).

IV. Concluding Remarks and Policy Implications

Though we focus on banks in the United States, due in part to it being the epicenter of the crisis and in part the better availability of banking data, evidence from Europe and the United Kingdom appears to be consistent with our findings.²⁷ Given that many large European banks were also exposed to the ABCP freeze starting in August 2007 (Acharya, Schnabl, and Suarez (2013)), the funding risks of these banks were similar to their U.S. counterparts (and in most cases worse, as these banks had limited access to the U.S. deposit market and government funding such as FHLB advances). For instance, there was (and remains) a depositor flight from the European periphery to stronger German banks (perceived to have a stronger government backing them). Similarly, while many British depositors increased their funds in Irish banks, drawn by higher rates after the Irish government full guarantee was announced in 2008 (roughly half of total deposits were held by nonresidents), several large Irish banks eventually experienced a deposit flight in November 2010. This shows how, when the aggregate shock risks the sovereign itself, the standard argument that banks function well as liquidity providers can fail due to the poor quality of deposit insurance.

Furthermore, during the 2007 to 2009 crisis, banks seem to have only partly helped avoid financial disruptions and business liquidations that would have occurred in the absence of a liquidity backstop by the central bank and the

²⁶ These specifications also include double interaction terms as a robustness check. The coefficients of interest are those on the triple interaction terms (see Bertrand, Schoar, and Thesmar (2007) for a similar triple interaction model). For example, before the recent crisis, an increase in the commercial paper (CP) spread by 100 basis points raised deposit growth 0.22 percentage points more at a bank with a high unused commitment ratio than at a bank with a low unused commitment ratio. In contrast, when market liquidity was highly stressed in the first phase of this crisis, deposit growth contracted 0.17 percentage points more at a bank with a high unused commitment ratio than at a bank with a low ratio.

²⁷ Two recent papers by Liu (2011) and Gozzi and Goetz (2010) also highlight the importance of funding shocks in crises. For example, Liu finds that banks with deposit exposure cut back lending significantly in branches outside the crisis country (funding shock), and the magnitude of this effect is twice as large as that stemming from nondepository asset exposure only (capital shock). Gozzi and Goetz find that, controlling for solvency position, a bank's reliance on wholesale funding was associated with reduced lending during the 2007 to 2009 crisis, and that employment fell in metropolitan areas with greater bank dependence on wholesale funding.

government. Our results suggest that the existing safety net was insufficient at sustaining a synergy between deposit-taking and commitment lending. This finding implies that policymakers need to design other—or alternative—mechanisms to ensure an uninterrupted supply of credit to creditworthy borrowers. Ex post, such mechanisms could include direct interventions in markets, such as the Federal Reserve's CPFF, which helped support issuance of short-term paper by businesses. Moreover, federal funding of temporary programs to backstop banks so as to avoid critical disruptions in credit and liquidity may be justified, provided they are suitably priced and made contingent on bank quality.

A part of this improvement could feature aligning deposit insurance premiums with still-existing implicit guarantees that encourage banks to take excessive risks.²⁸ Pennacchi (2006) and Acharya, Santos, and Yorulmazer (2010) argue that the premium charged should not just be actuarially fair (so the fund breaks even on average) but also reflect an additional component to discourage moral hazard. The FDIC, following the Dodd-Frank Act of 2010, proposed some new reforms in this direction to make deposit insurance premiums more sensitive to the systemic risk of an institution and not just to its idiosyncratic risk. Our results show, however, that the liquidity risk of institutions (for instance, the risk of drawdowns on promised credit lines) may also be a relevant factor, and that liquidity risk interacts with the fundamental or solvency risk of institutions.

On the liquidity risk front, better liquidity regulation is being proposed under Basel III in the form of a Liquidity Coverage Ratio (LCR) and a Net Stable Funding Ratio (NSFR). More progress to date has been made on the LCR, which requires banks to hold a sufficient amount of unencumbered high-quality liquid assets to cover funding outflows over 30 days in a liquidity stress event. The rationale for an ex ante LCR is that ex post intervention in the form of the lender-of-last-resort by the central bank can be socially costly (e.g., Rochet and Vives (2004), Diamond and Rajan (2005), Stein (2013)). First, in reality, even central banks cannot perfectly distinguish between illiquid and insolvent banks so that liquidity guarantees come with credit risk incurred by taxpayers. Second, the use of government backstops in a crisis induces a moral hazard problem. Therefore, as Stein (2013) argues, if access to the central bank's liquidity facilities is to count toward an LCR requirement, it makes sense to price this access (e.g., an up-front fee on a loan commitment from the central bank, as done in Australia).

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²⁸ Indeed, risk-shifting may be exacerbated if the heightened competition for deposits during a bank-centered crisis further lowers bank charter values and induces more risk-taking on the asset portfolio when combined with deposit insurance (see Keeley (1990)).

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Internet Appendix.